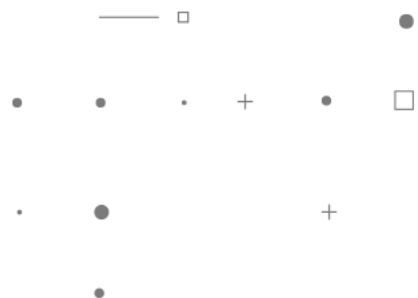


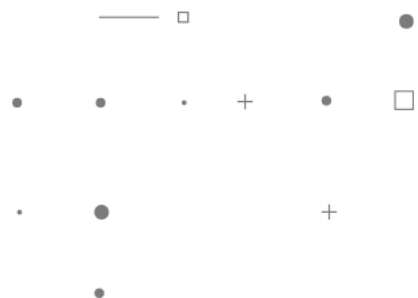
FIAP

NABA



MBA em Engenharia de Dados





Graph Database & Analytics





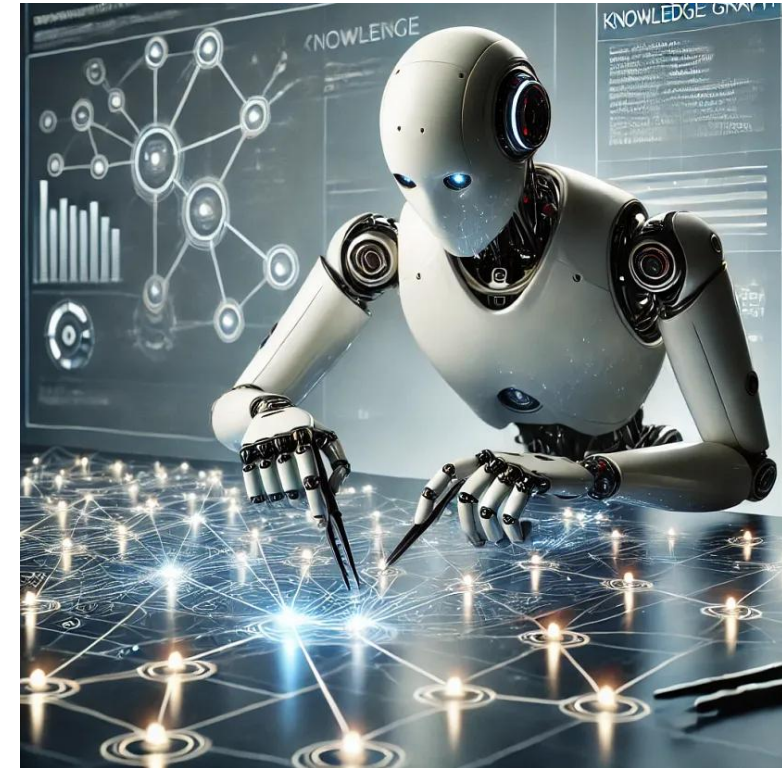
Graph Data Science

Knowledge Graphs

✓ LLMs e KG



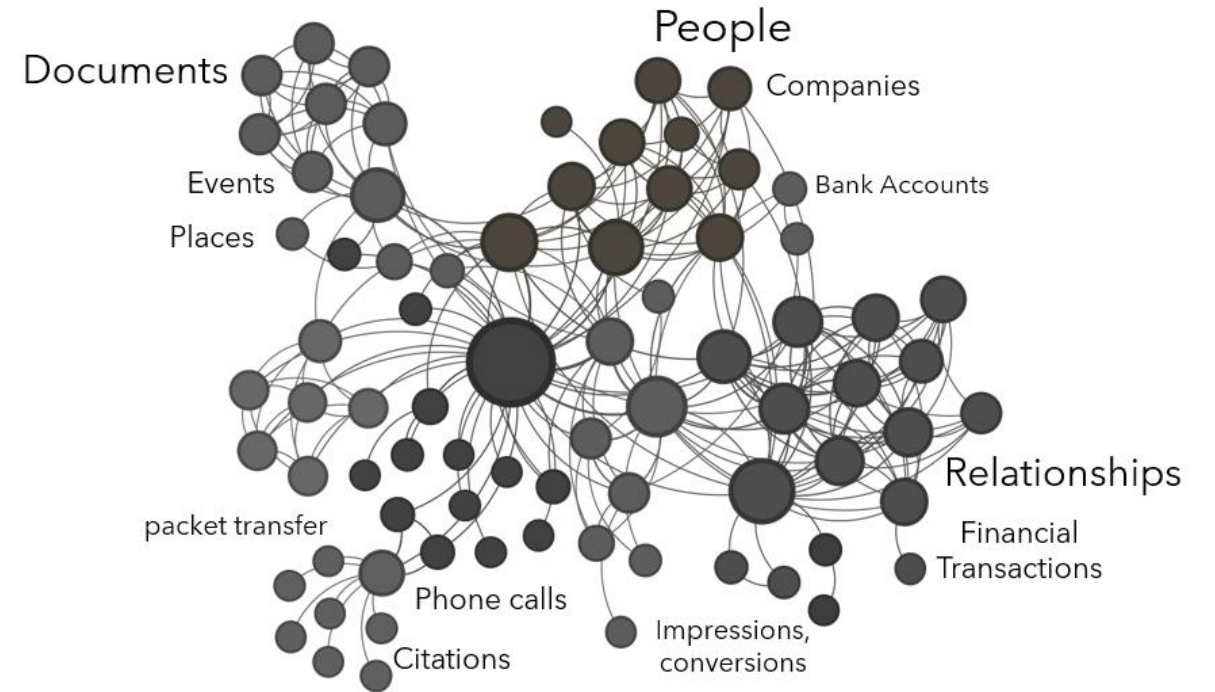
Knowledge Graph é
sobre conectar
conhecimento.





Knowledge Graph

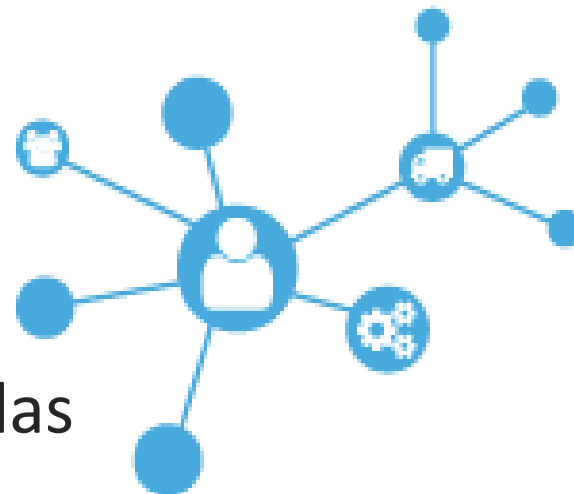
FIAP





Knowledge Graph

- Define entidades do mundo real
- Providencia relacionamento entre elas
- Contem regras definidas por meio de ontologias
- Habilita o raciocínio para inferir novos conhecimentos simulando o raciocínio lógico dedutivo



Google Knowledge Graph (maio 2012)



Knowledge Graph

“Base de conhecimento semântica que organiza fatos sobre pessoas, lugares e coisas e entende as relações entre eles para entregar resultados de busca mais contextuais e relevantes, alimentando os painéis de conhecimento e respostas diretas na pesquisa.



Google Knowledge Graph

Existem “*things*” não “*strings*”

Engenheiro de Dados sênior em São Paulo



Título da Vaga

Nível

Localização



Google Knowledge Graph

FIAP



Elon Musk
CEO of SpaceX

Elon Reeve Musk is a South African-born American business magnate, investor, and engineer. He is the founder, CEO, and lead designer of SpaceX; co-founder, CEO, and product architect of Tesla, Inc.; and co-founder and CEO of Neuralink. [Wikipedia](#)

Born: June 28, 1971 (age 46 years), Pretoria, South Africa
Net worth: 19.3 billion USD (2018) [Forbes](#)
Education: University of Pennsylvania (1997), [MORE](#)
Parents: [Maye Musk](#), [Errol Musk](#)
Spouse: [Talulah Riley](#) (m. 2013–2016), [Talulah Riley](#) (m. 2010–2012), [Justine Musk](#) (m. 2000–2008)
Children: [Nevada Alexander Musk](#), [Kai Musk](#), [Xavier Musk](#), [Damian Musk](#), [Saxon Musk](#), [Griffin Musk](#)

Profiles

[Twitter](#) [Instagram](#) [Google+](#)

People also search for [View 15+ more](#)

[Talulah Riley](#)
Former spouse

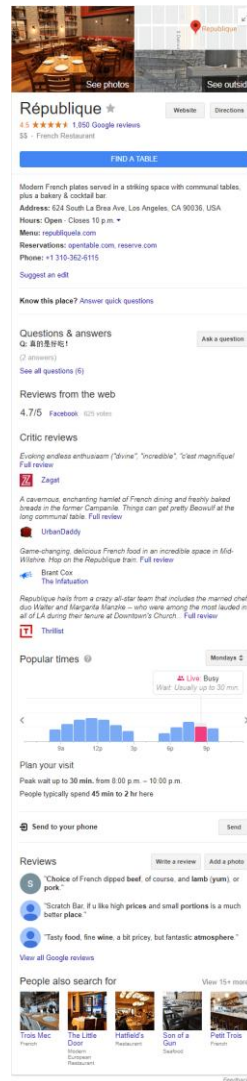
[Tim Cook](#)

[Bill Gates](#)

[Justine Musk](#)
Former spouse

[Jeff Bezos](#)

[Feedback](#)



République ★
4.5 ★★★★★ 1,650 Google reviews
55 French Restaurant

[Find a table](#)

Modern French plates served in a dining space with communal tables, plus a bakery & cocktail bar.
Address: 624 South La Brea Ave, Los Angeles, CA 90036, USA
Hours: Open · Closes 10 p.m. ·
Menu: [republiquela.com](#)
Reservations: [opentable.com](#), [reservec.com](#)
Phone: +1 310-362-6115
[Suggest an edit](#)

Know this place? Answer quick questions

Questions & answers
2 answers
[See all questions \(6\)](#)

Reviews from the web
4.7/5 [Facebook](#) 625 votes

Critic reviews
[Eating and drinking](#) (3/4) "Incredible", "best magnificent" Full review
[Zagat](#)
A cavernous, enchanting hamlet of French dining and freshly baked breads in the former Campanile. Things can get pretty busy at the long communal table. Full review
[UrbanDaddy](#)
Game-changing, delicious French food in an incredible space in Mid-Wilshire. Hop on the République train. Full review
[Bart Cox](#) The Information
République hails from a crazy all-star team that includes the married chef duo Walter and Margarita Blázquez – who were among the most lauded in all of LA during their tenure at Döner's Church. Full review
[Thriller](#)

Popular times @
Monitors 2
Live Busy
Wait: Usually up to 30 min

Plan your visit
Peak wait up to 30 min, from 6:00 p.m. – 10:00 p.m.
People typically spend 45 min to 2 hr here

[Send to your phone](#) [Send](#)

Reviews [Write a review](#) [Add a photo](#)

1 "Choice of French dipped beef, of course, and lamb (yam), or pork."
2 "Scratch Bar, if u like high prices and small portions is a much better place."
3 "Tasty food, fine wine, a bit pricey, but fantastic atmosphere."

[View all Google reviews](#)

People also search for [View 15+ more](#)

[Tru Mee](#) [The Little Door](#) [Pattino's](#) [Sun of a Gun](#) [Pett Toss](#)
French Italian American Italian American French

[Feedback](#)

- ✓ Opinião dos consumidores.
- ✓ Revisões críticas.
- ✓ Horas diárias.
- ✓ Tempos populares.
- ✓ Opções de reserva.
- ✓ Números de telefone.
- ✓ Instruções.
- ✓ Um link para o seu site.
- ✓ E mais.

Google Knowledge Graph

FIAP

Avengers: Infinity War

2018 · Fantasy/Science fiction film · 2h 40m

[Play trailer on YouTube](#)

9.7/10 · IMDb

97% liked this film
Google users

Iron Man, Thor, the Hulk and the rest of the Avengers unite to battle their most powerful enemy yet -- the evil Thanos. On a mission to collect all six Infinity Stones, Thanos plans to use the artifacts to inflict his twisted will on reality. The fate of the planet and existence itself has never been...

Release date: April 27, 2018 (Canada)
Directors: Anthony Russo, Joe Russo
Budget: \$300-\$400 million
Music composed by: Alan Silvestri
Film series: The Avengers film series

Cast [View 15+ more](#)



Robert Downey Jr.
Iron Man



Chris Pratt
Star-Lord



Chris Evans
Captain America



Josh Brolin
Thanos



Chris Hemsworth
Thor

People also search for [View 120+ more](#)



The Avengers
2012



Black Panther
2018



Avengers: Age of Ultron
2015

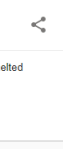
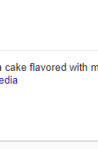
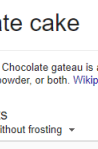
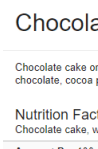


Deadpool 2
2018



Captain America: Civil War
2016

[Feedback](#)



[More images](#)

Chocolate cake

Chocolate cake or Chocolate gâteau is a cake flavored with melted chocolate, cocoa powder, or both. [Wikipedia](#)

Nutrition Facts
Chocolate cake, without frosting

Amount Per 100 grams

	% Daily Value*
Calories 371	
Total Fat 15 g	23%
Saturated fat 5 g	25%
Polyunsaturated fat 2.8 g	
Monounsaturated fat 6 g	
Cholesterol 58 mg	19%
Sodium 315 mg	13%
Potassium 140 mg	4%
Total Carbohydrate 53 g	17%
Dietary fiber 1.6 g	6%
Protein 5 g	10%
Vitamin A	2%
Vitamin C	0%
Calcium	6%
Iron	8%
Vitamin D	0%
Vitamin B-6	0%
Vitamin B-12	3%
Magnesium	8%

*Per cent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower depending on your calorie needs.

People also search for [View 15+ more](#)



Chocolate



Chocolate brownie



Birthday cake



Icing



Cheesecake

Sources include: USDA

[Feedback](#)

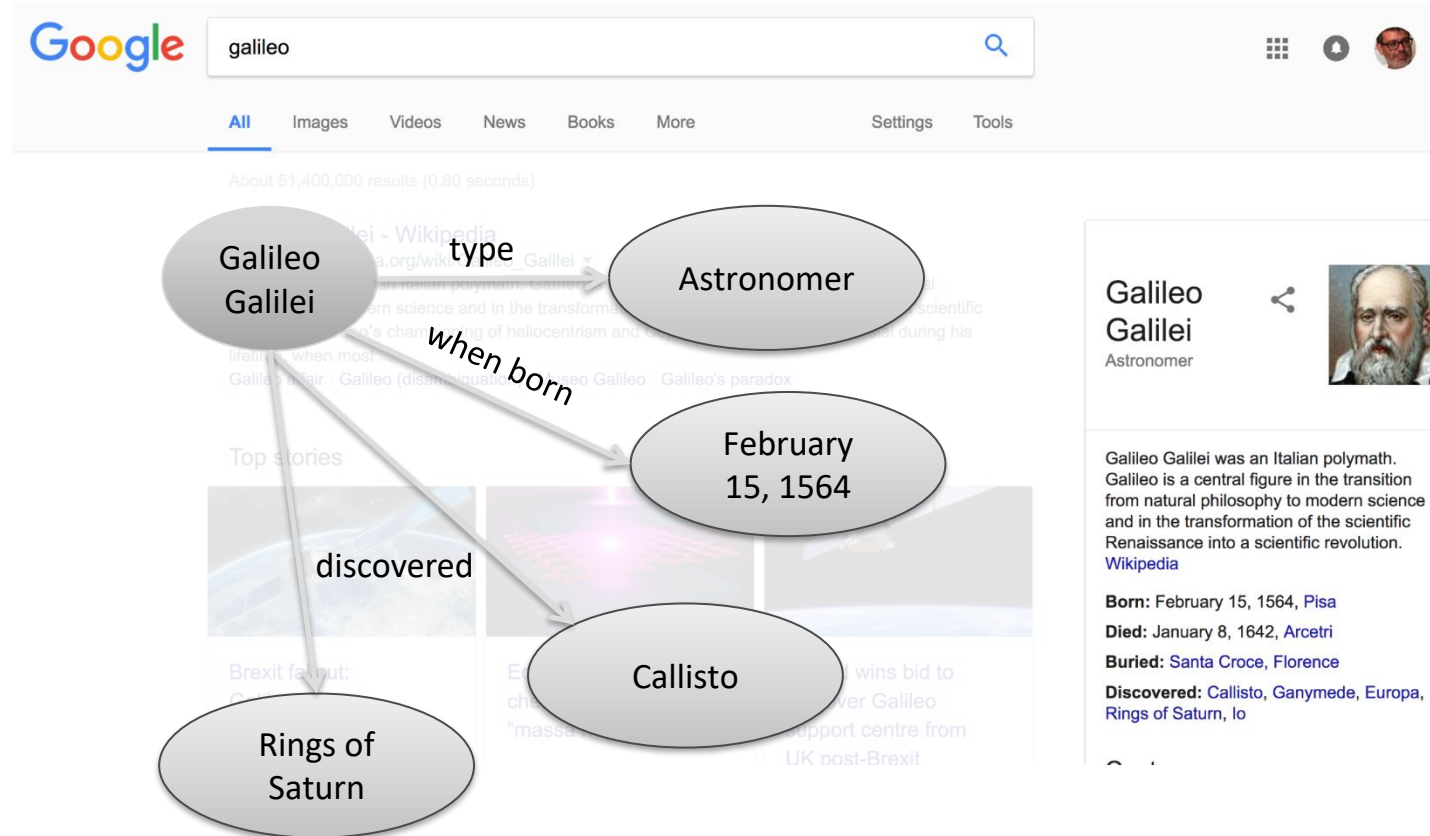
✓ Marcação schema.org

✓ Wikidata e wikipedia

✓ Google maps

✓ Contas nas mídias sociais

Google Knowledge Graph





Vamos pesquisar sobre “puma” numa
plataforma de busca?

RECURSOS SEM ANOTAÇÃO X ANOTADOS





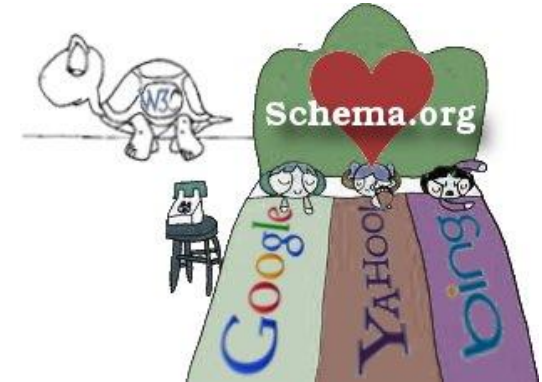
“Metadata is data about data”

“Dados estruturados sobre dados”

Exemplos:

- Significado para criação do recurso
- Propósito do recurso
- Data e horário de criação
- Autor
- Localização na rede onde o dado foi criado
- Padrões são usados

schema.org



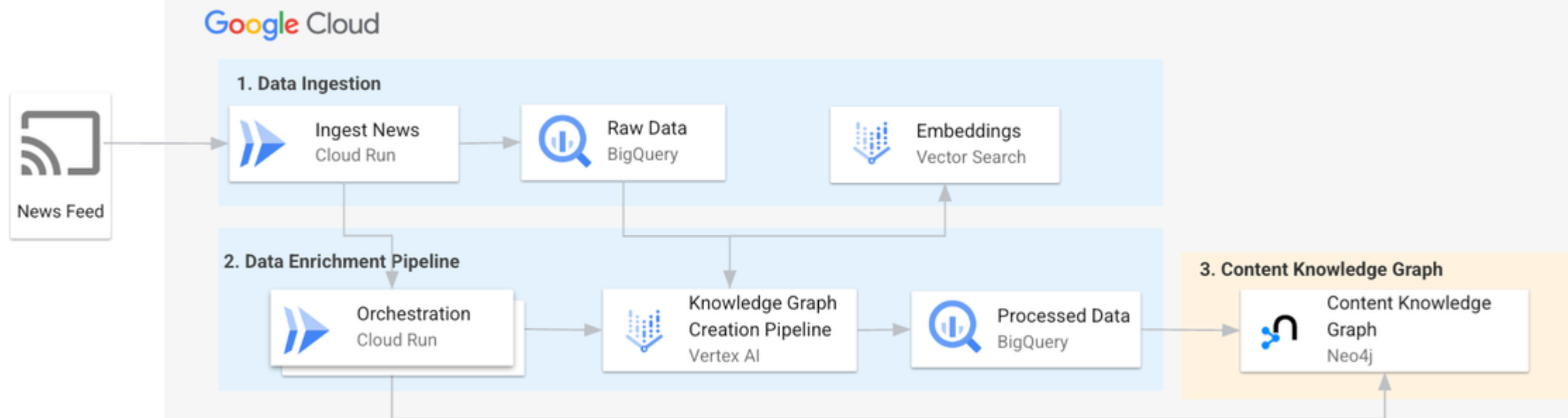
- Atividade colaborativa com a missão de criar, manter e promover metadados - esquemas para dados estruturados - para recursos na web como páginas, mensagens de e-mail, vídeos, entre outros.
- *2010 / 2011*

Google Knowledge Graphs

Case Glance

Utiliza o Google Cloud para construir um Grafo de Conhecimento de Conteúdo (CKG) com tecnologia Gemini.

Architecture: Content Knowledge Graph

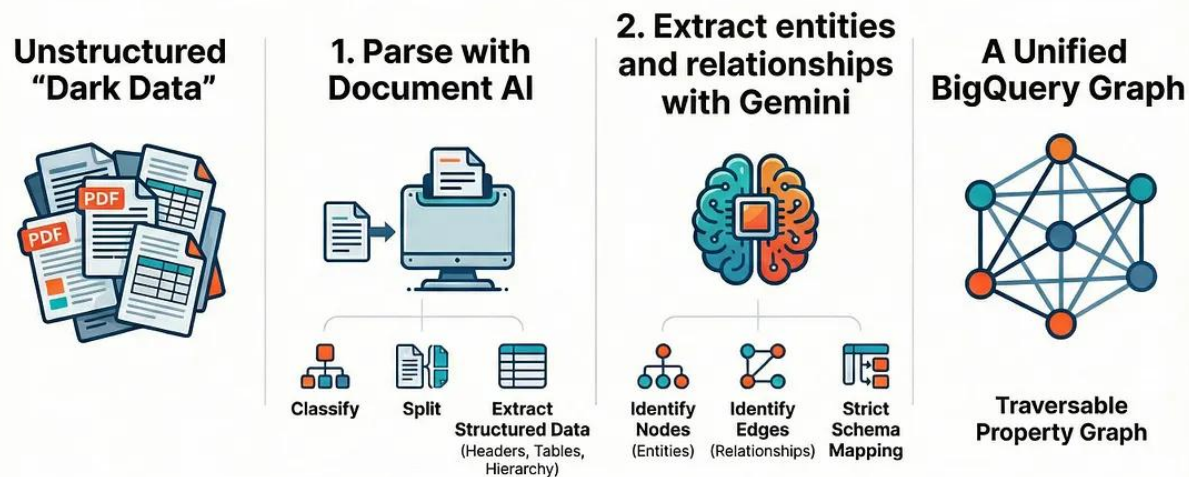


Google Knowledge Graphs

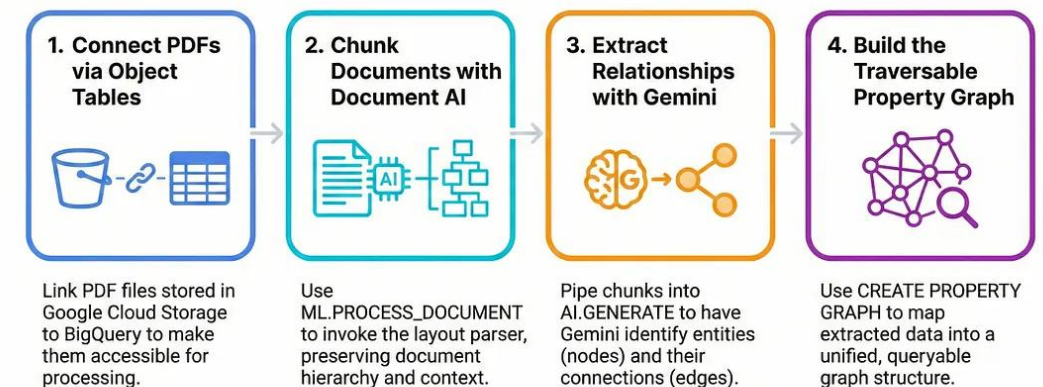


Big Query Graphs

Hybrid AI: From PDFs to Knowledge Graphs



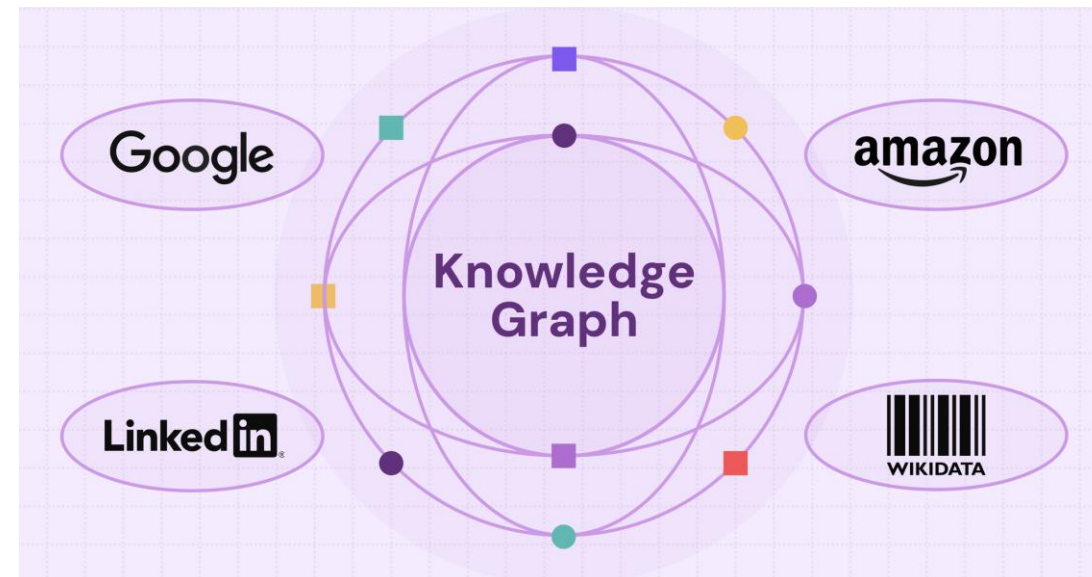
BigQuery Graph Pipeline



Knowledge Graph

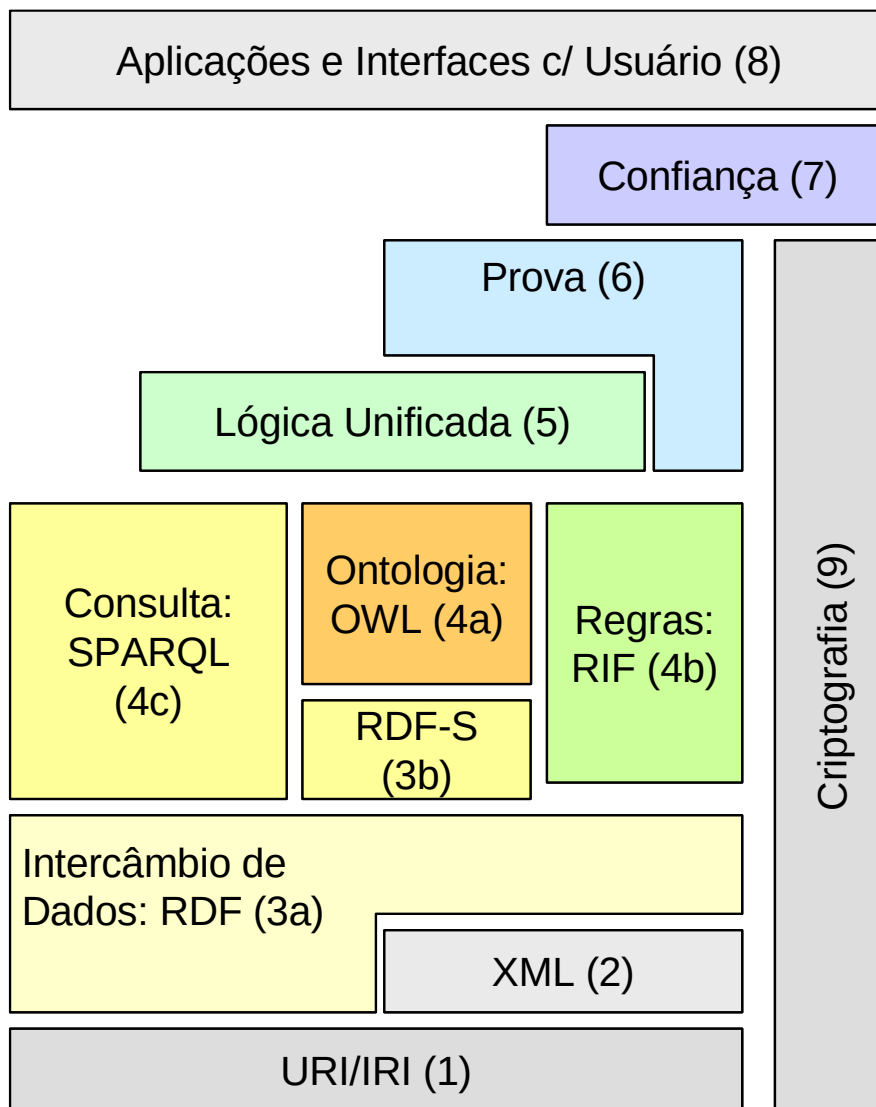
Neo4j Knowledge Graph Leaders

Recordar slide



- Google Knowledge Graph
- Google Knowledge Vault
- Microsoft Satori
- Amazon Product Graph
- Facebook Graph API
- IBM Watson
- LinkedIn Knowledge Graph

Knowledge Graphs



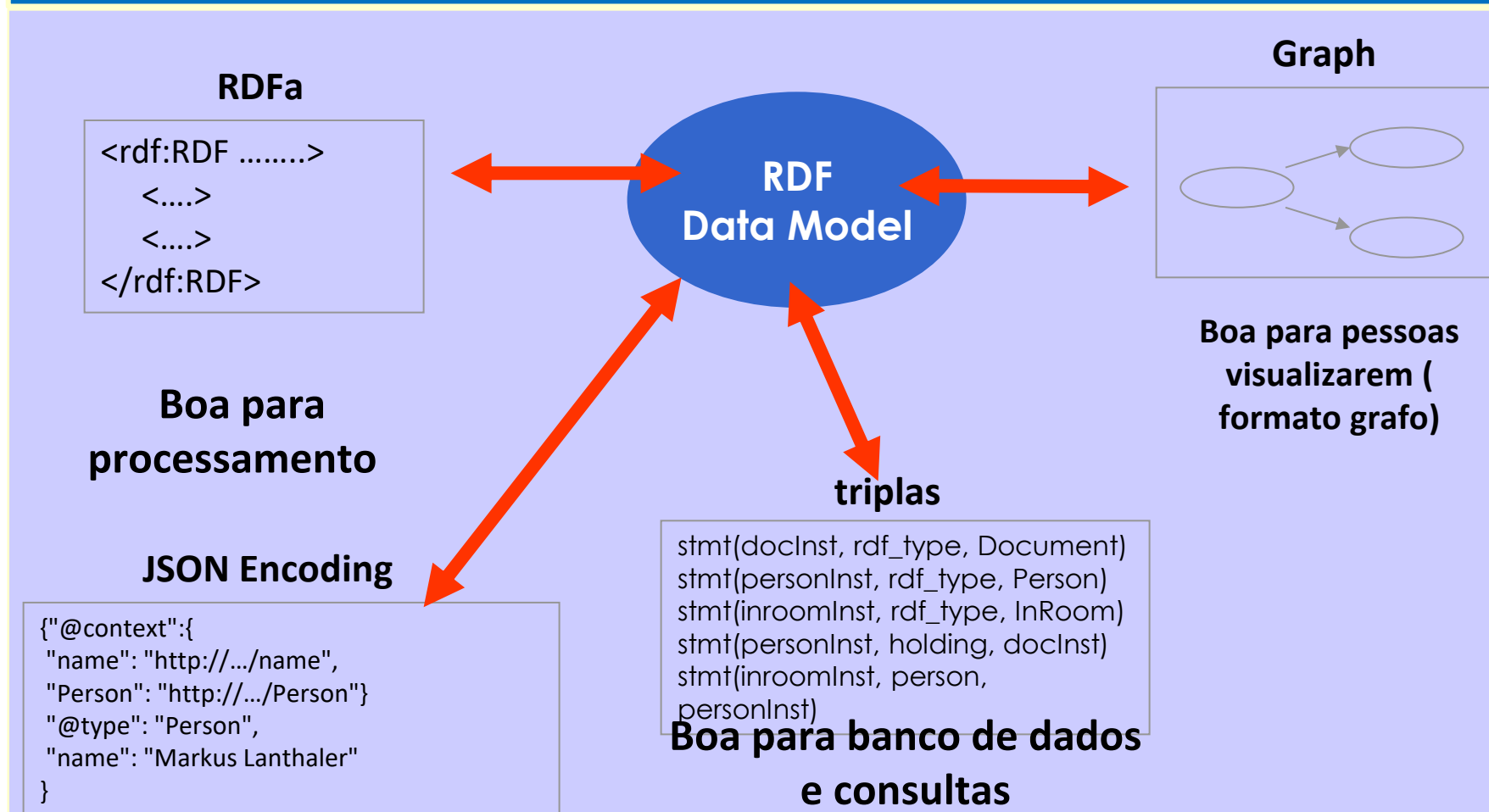
Arquitetura proposta W3C

Arquitetura W3C, em suas camadas, define as tecnologias necessárias para que os conteúdos das páginas Web possam ser compreendidos pelos computadores.

RDF (*Resource Description Framework*)



RDF é uma linguagem simples para representar informação sobre recursos existentes na web baseadas em grafos





RDF

Linked Open Data (LOD)





SPARQL

FIAP



- Linguagem de consulta para dados RDF, permitindo a recuperação de informação contida em grafos
- Desenvolvida pela W3C em 2008
- Protocolo de acesso a dados RDF

SPARQL Protocol and RDF Query Language

SPARQL

Partes de uma consulta SPARQL:

Declaração de Prefixos

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
PREFIX foaf: <http://xmlns.com/foaf/0.1/>  
PREFIX dbpedia-owl: <http://dbpedia.org/ontology/>
```

```
SELECT ?nomeUniversidade
```

Cláusula de Resultado

```
FROM <http://dbpedia.org>
```

Grafos RDF Consultados

```
WHERE
```

```
{
```

```
    ?universidade rdf:type dbpedia-owl:University .
```

```
    ?universidade foaf:name ?nomeUniversidade .
```

```
    ?universidade dbpedia-owl:country <http://dbpedia.org/resource/Brazil> .
```

```
}
```

Padrão de Consulta

```
ORDER BY ?nomeUniversidade
```

Modificadores de Consulta



RDF - SPARQL

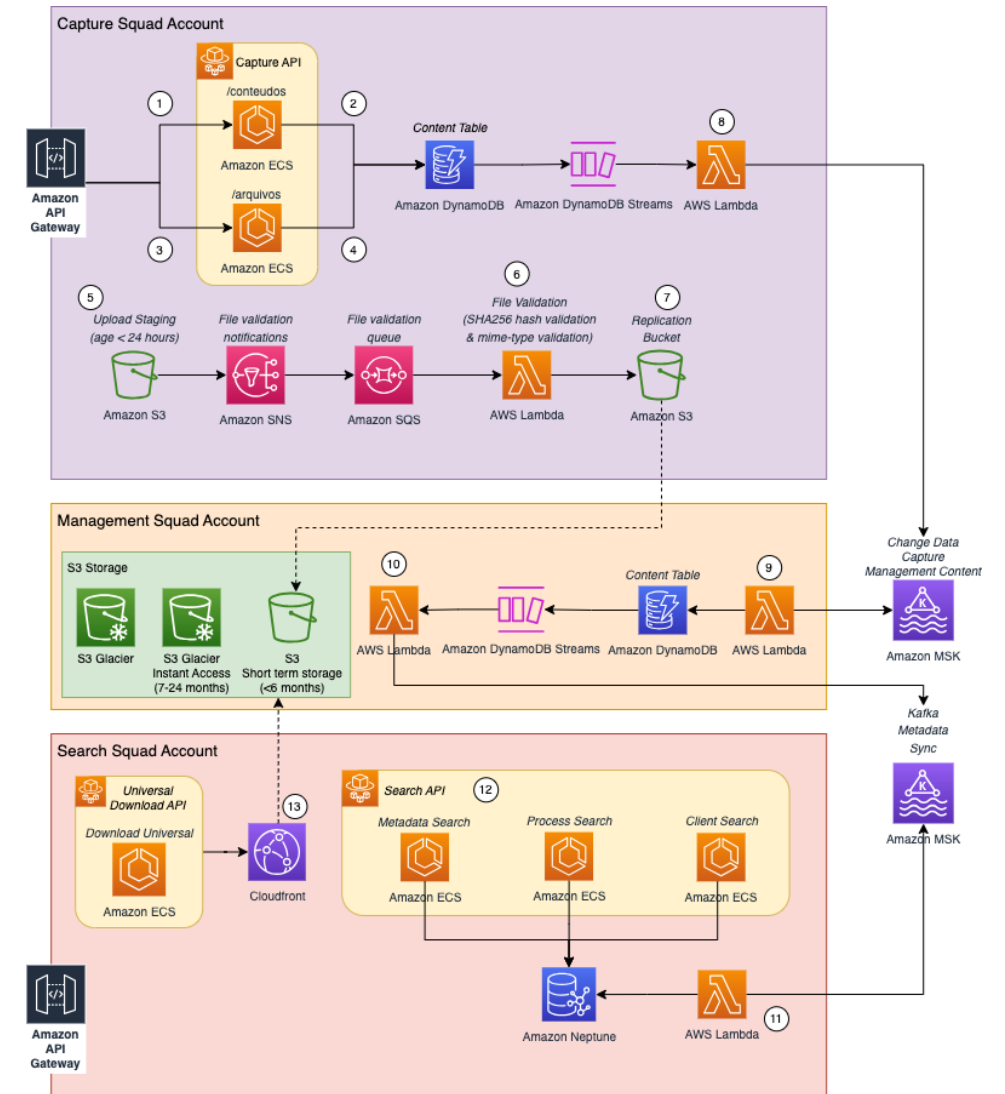
Names of artists who have had albums produced by Irving Townsend?



prefix schema: <<http://schema.org/>> .

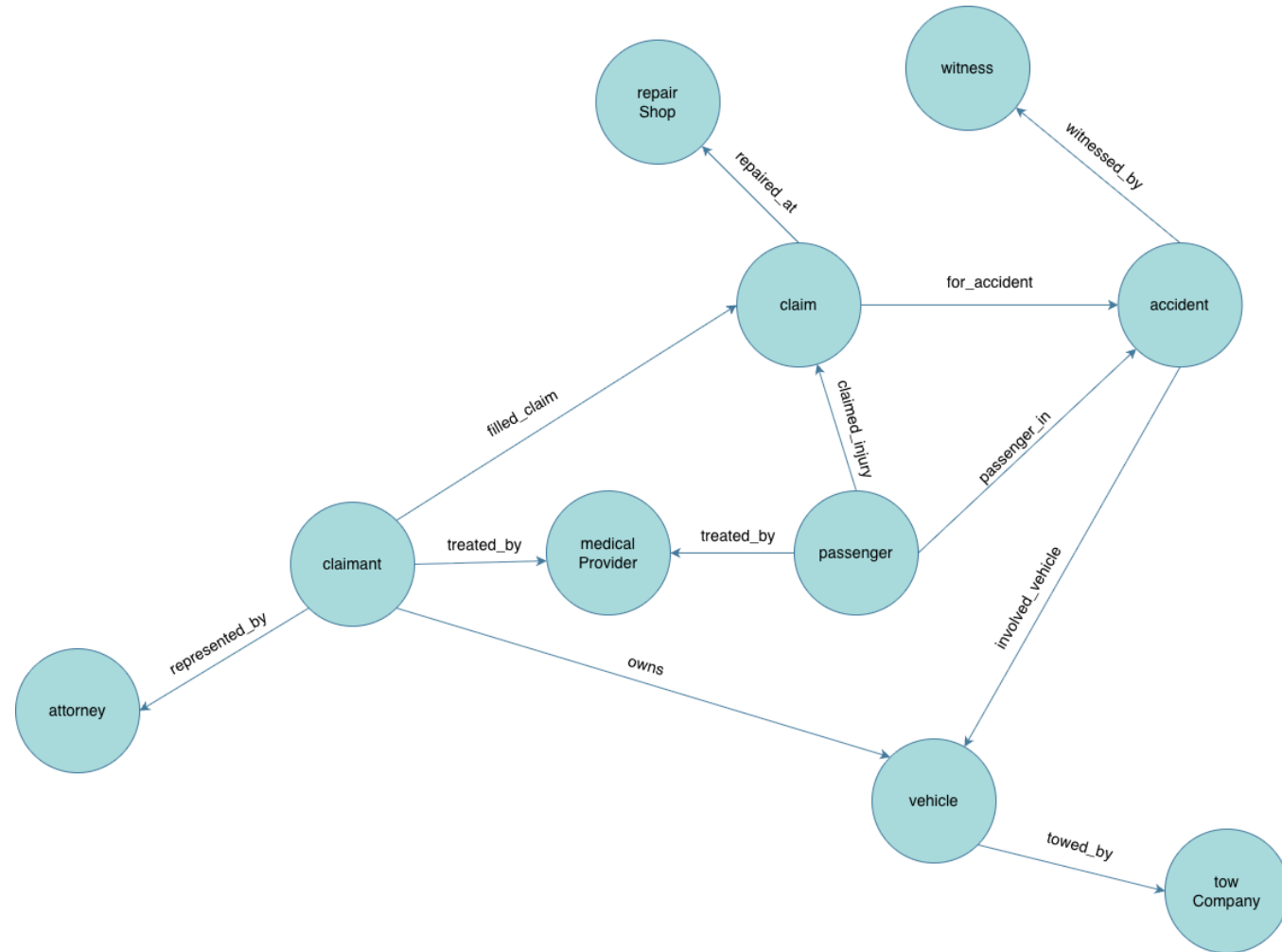
```
SELECT DISTINCT ?artistname
WHERE {
```

```
?prod schema:name "Irving Townsend"
?musalb schema:producer ?prod .
?musalb schema:byArtist ?artist .
?artist schema:name ?artistname .
```



Knowledge Graph

AWS Neptune
Detecção de Fraude



Exemplo

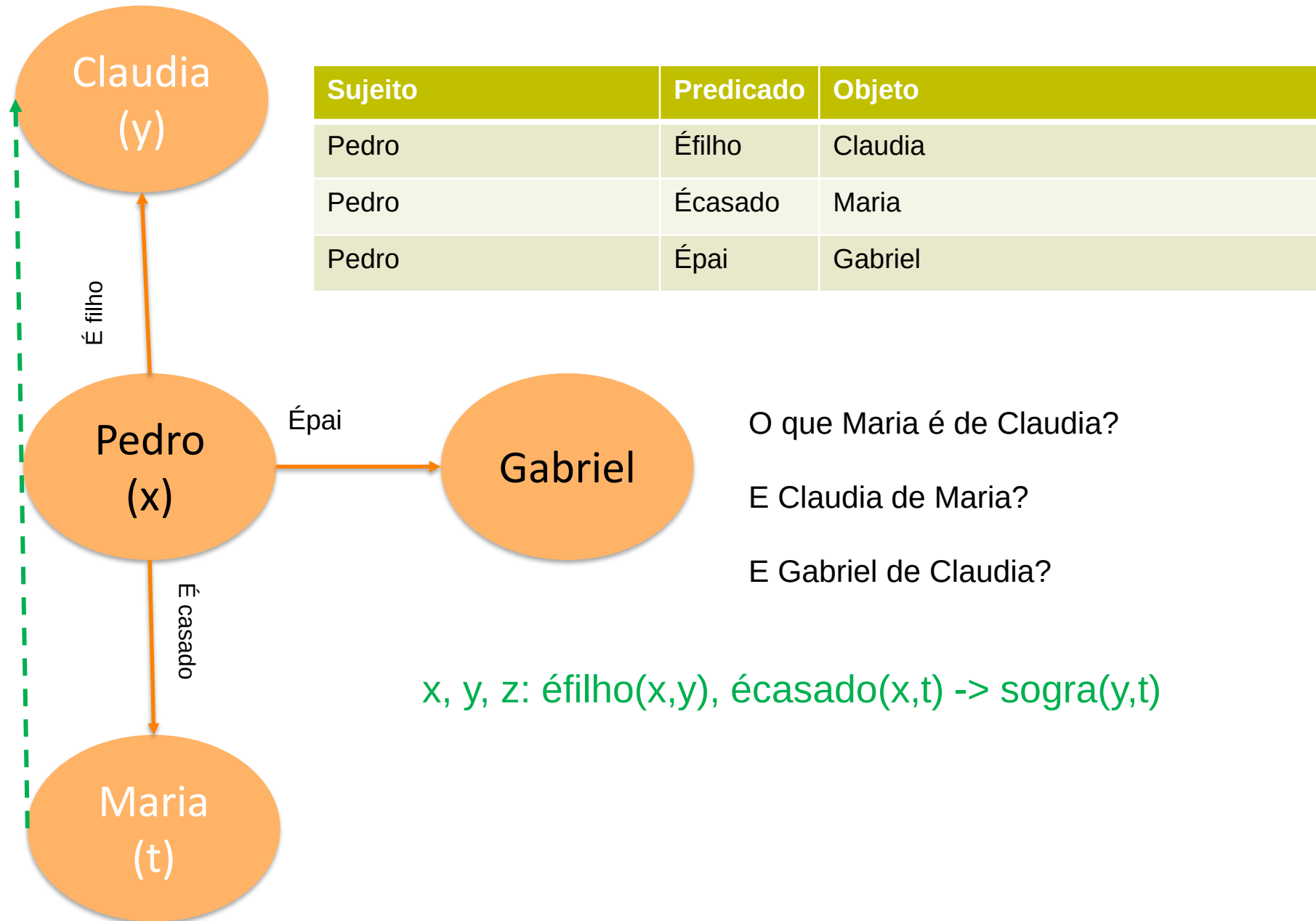
(1) Realizar as consultas propostas em:

a) <https://query.wikidata.org/>



Inferência

Knowledge Graphs - Inferência

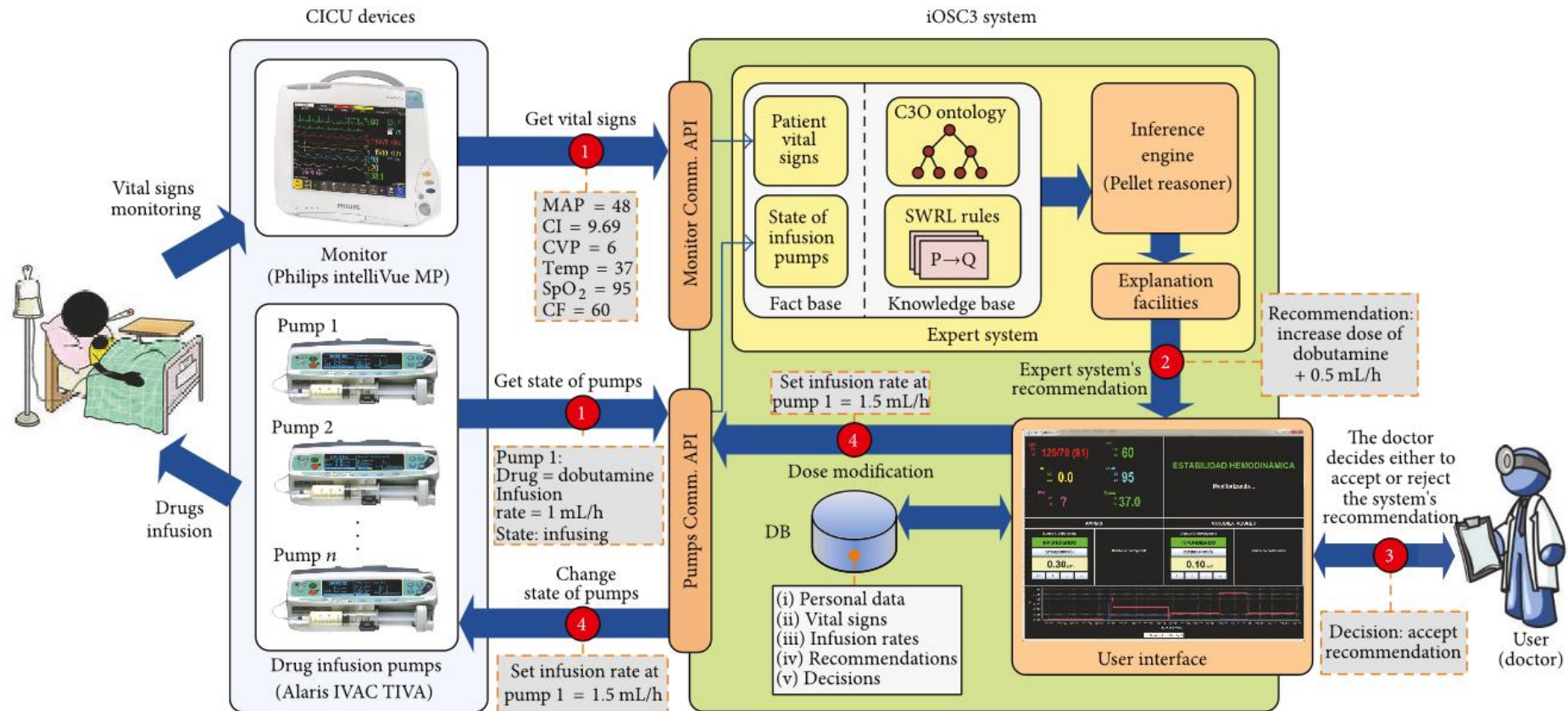


Knowledge Graphs - Inferência

$\forall x, y : atacante(x) \wedge chute(x, y) \wedge não-defende(y) \rightarrow gol(x)$



Knowledge Graphs





Knowledge Graphs

Reasoners

Motor de inferência

Um programa que utiliza as regras em combinação com os fatos para gerar novos fatos.



Exemplo:

<https://docs.stardog.com/inference-engine/user-defined-rules>

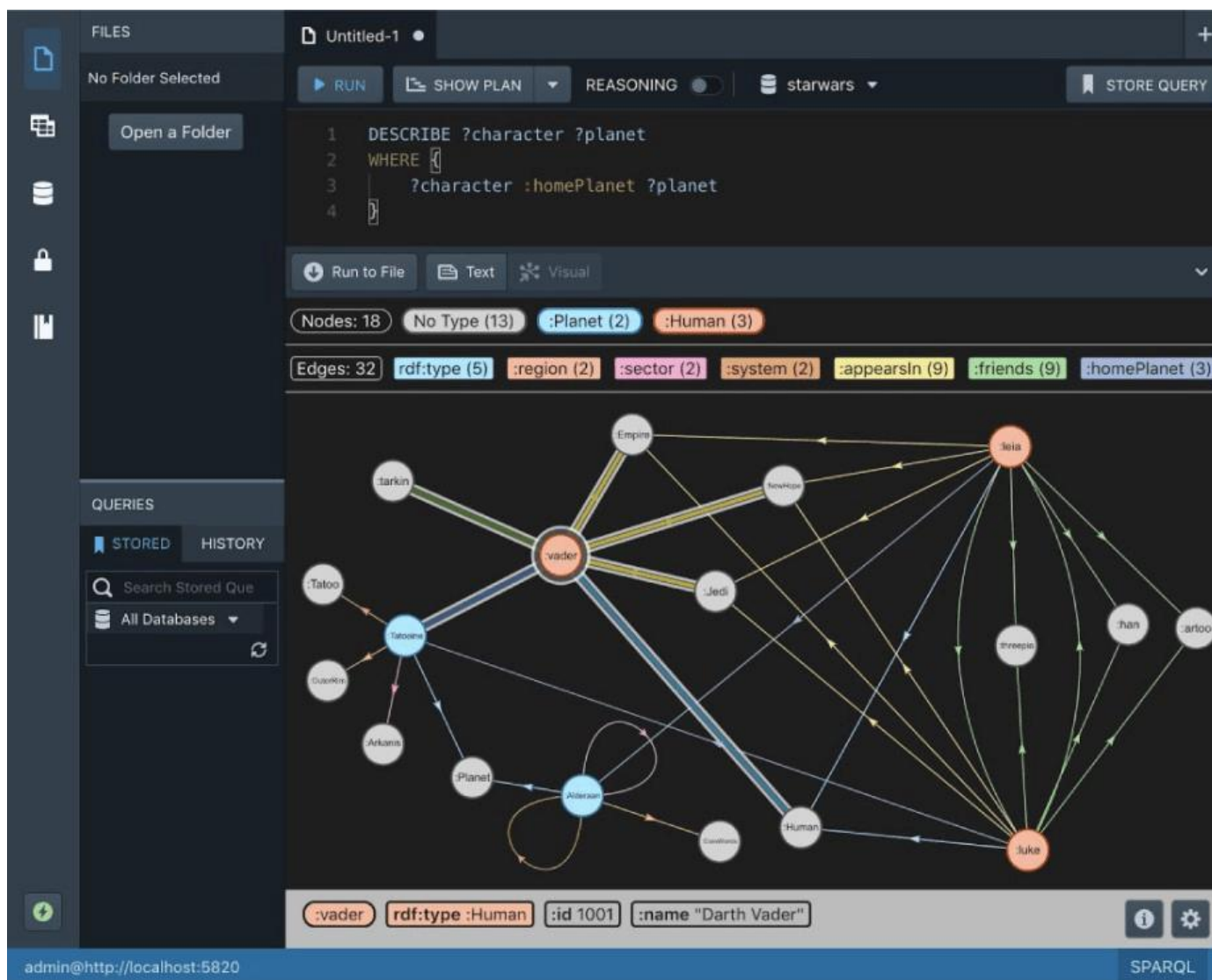


Stardog

- <http://stardog.com/>
- Banco de dados RDF puro Java (“quad store”)
- Muito rápido para uso in-memory
- Reasoner suporta via Pellet para OWL DL e consultas para OWL 2 QL, EL e RL
- Interface comando de linha e JAVA API

Knowledge Graph

FIAP



<https://cloud.stardog.com/kits/>



Knowledge Graphs



Select the plan that works for you

Free

Learning environment to create your first Knowledge Graph.

Free

Feature Highlights

- ✓ Full Stardog EKG Platform
- ✓ Store up to 10M edges in 3 databases
- ✓ Community Support, no SLA

Get Stardog Free

Essentials

Everything you need for a PoC or development environment.

\$99/month

Everything in Free, plus

- ✓ Store up to 50M edges
- ✓ 5 Databases
- ✓ Community Support, 95% uptime SLA

Get Stardog Essentials

Enterprise

Full production environment tailored to the needs of your company.

Custom

Everything in Essentials, plus

- ✓ Unlimited edges
- ✓ Unlimited databases
- ✓ Premium Support, 99.9% uptime SLA

Contact Us

Demos

Learn Stardog basics by browsing industry-related and pop culture Knowledge Graphs.

[Finance] Insurance Underwriting Risk

A demo about insurance underwriting risk and exposure; see if you can find 'Acme Hardware'.

Reasoning

Insurance Underwriting 360

Ask questions across multiple policy admin and claims management systems to evaluate a request for a new home owners policy.

Financial_Services Insurance Reasoning

Experimental Research Protocol

A small kit representing an experimental protocol from bio-protocol.org and a non-experimental clinical protocol from atlas-medical as RDF.

Life_Sciences SPARQL Studio

Supply Chain 360

Ask questions across your manufacturing, suppliers, distributors and customer applications to investigate possible root cause of customer feedback about flat beer.

Explorer Manufacturing SPARQL Studio

Customer 360

Ask questions across your sales, marketing and service applications to get a complete and comprehensive understanding of your accounts, contacts, products, campaigns, interactions and support cases.

KG Training: C360

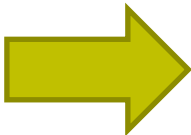
A simple customer c360 example dataset used for training purposes.

Medical History

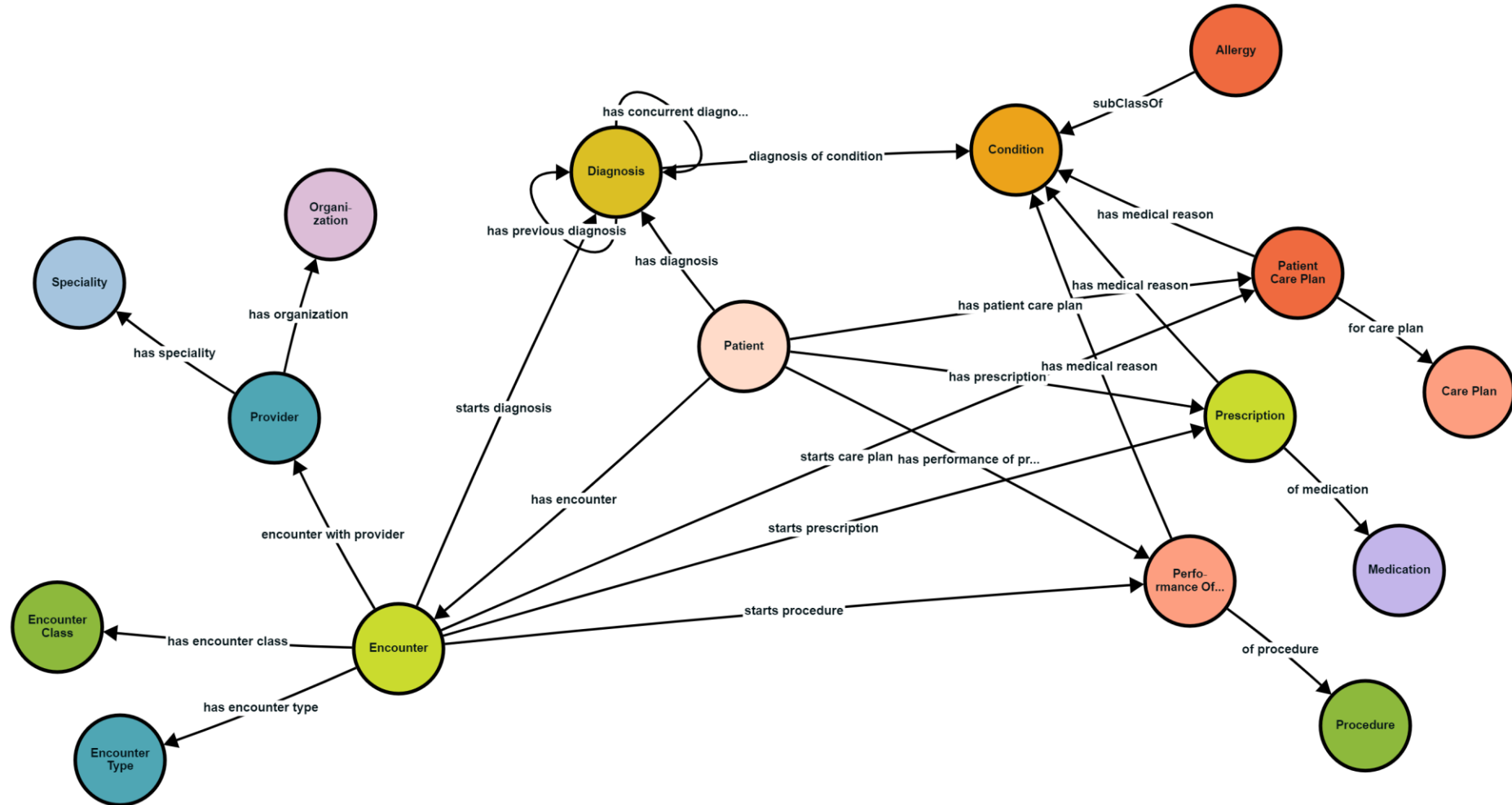
A healthcare demo that includes a small synthetic data set.

Flight Planning

An aviation focused dataset and flight planning demo



Knowledge Graphs



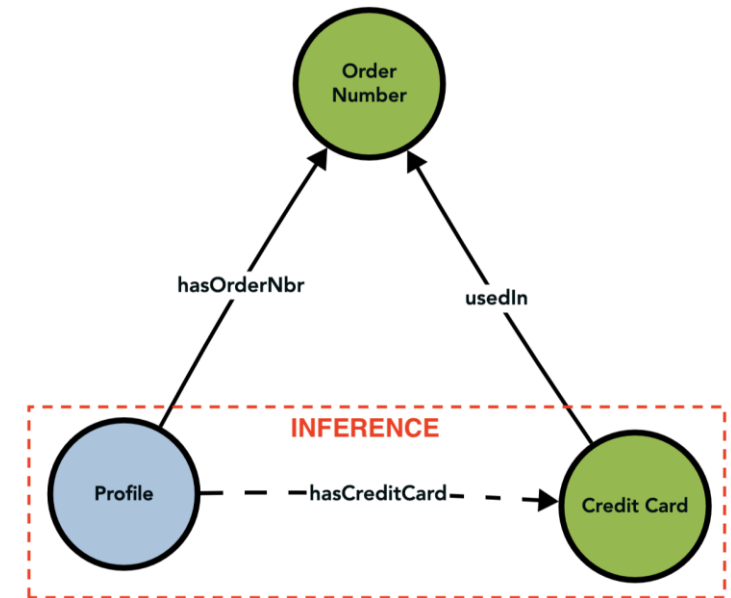
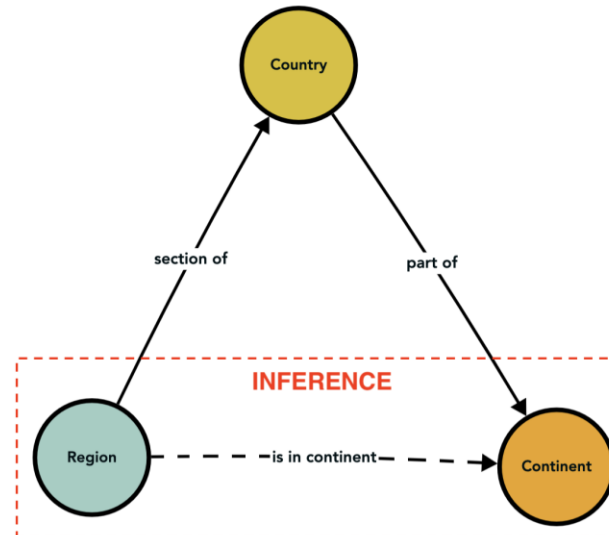
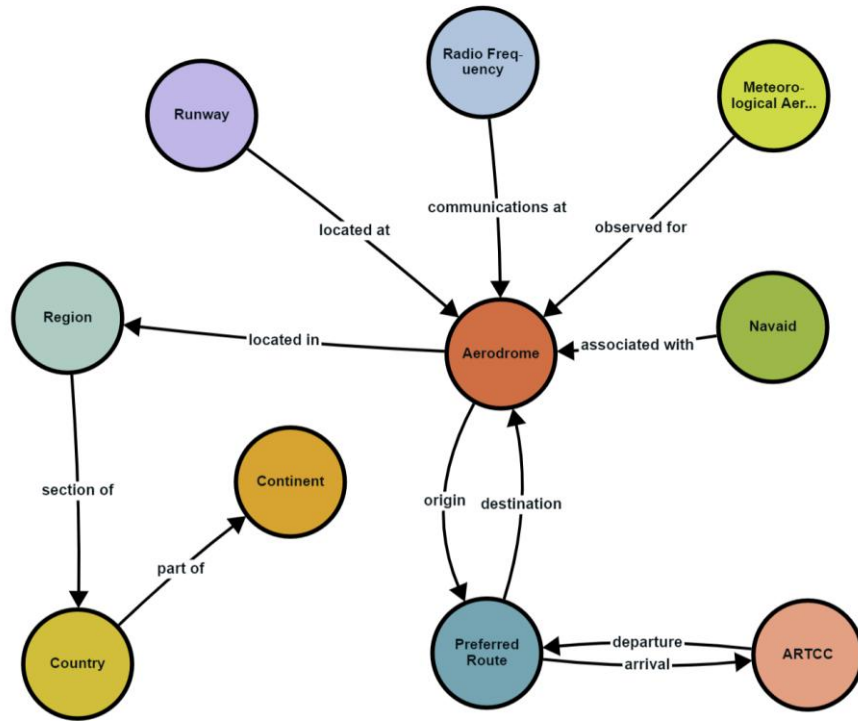


StarDog



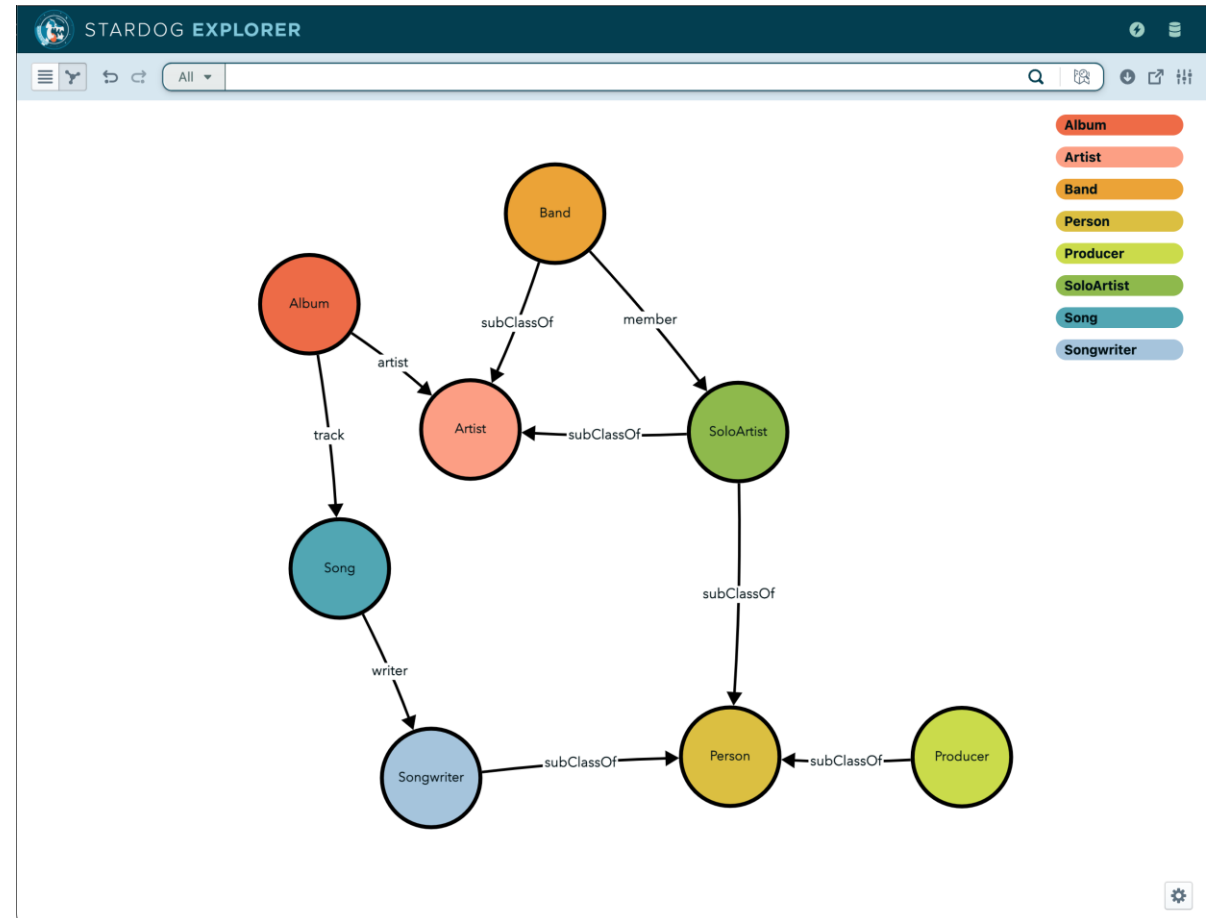
Open this demo in Explorer to get started:

[Open in Explorer](#)



StarDog

Tutorial Music



LLMs e Knowledge Graph

- Na era da IA, seu banco de dados não é apenas uma escolha de backend — é um facilitador estratégico. Os sistemas de IA não estão apenas consumindo dados, estão raciocinando e inferindo dados.

Na pilha nativa de IA, bancos de dados vetoriais e de grafos são fundamentais:

Os vetoriais são o núcleo da GenAI - tornam possíveis buscas semânticas, embeddings e arquiteturas RAG.

Os grafos são ideais para modelar relacionamentos, raciocínio e alimentar grafos de memória e decisão de agentes.

LLMs recuperam pedaços semanticamente correspondidos via busca vetorial.

Agentes raciocinam por meio de travessias de grafos e caminhos de decisão.

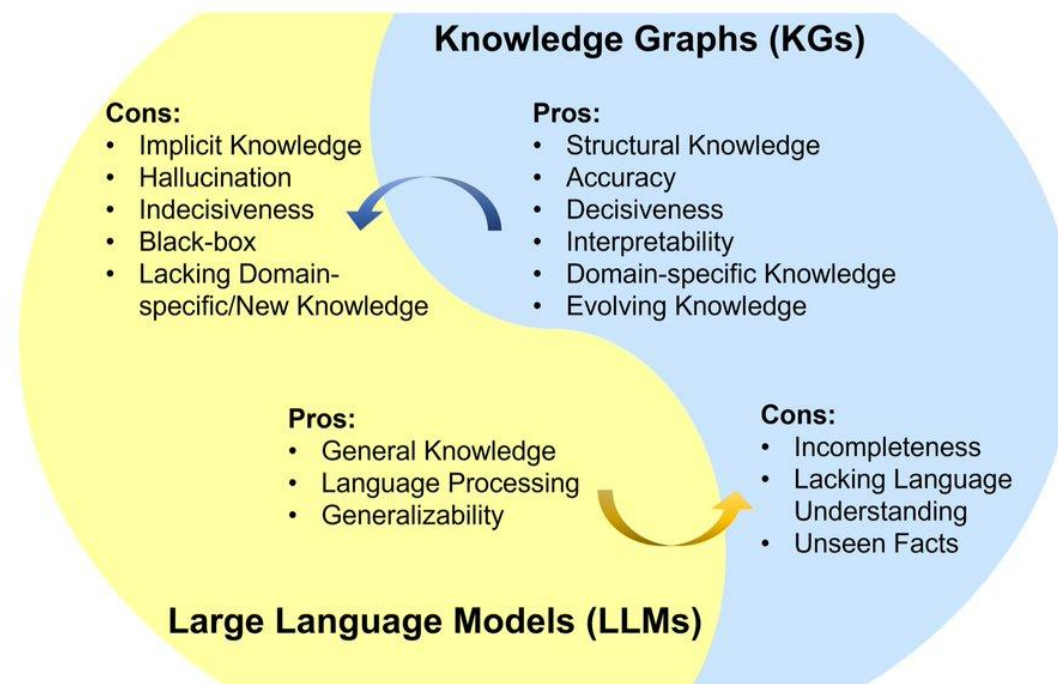
Modelos híbridos recuperam via vetor e raciocínio via grafo.

Não se trata apenas de armazenamento de dados — trata-se de possibilitar inteligência:

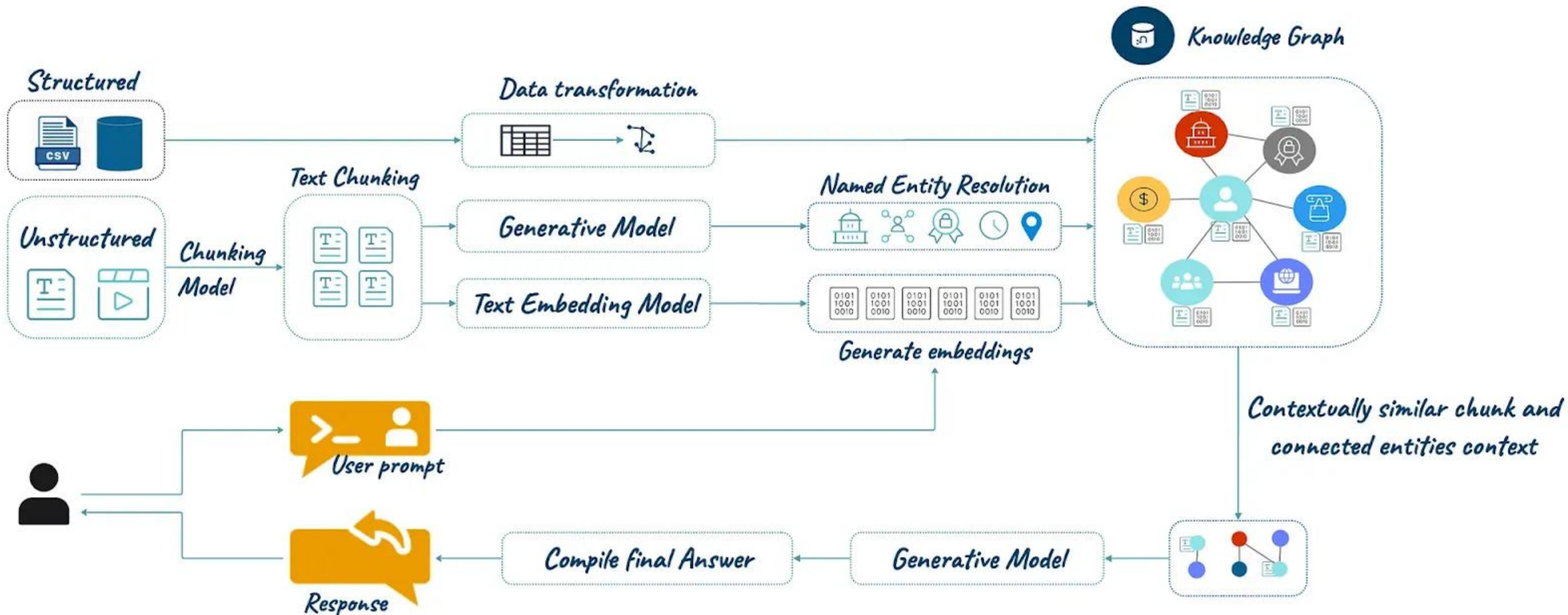
LLMs e Knowledge Graph

LLMs e grafos de conhecimento (KGs) são tecnologias complementares que equilibram os pontos fortes e fracos uns dos outros quando combinados:

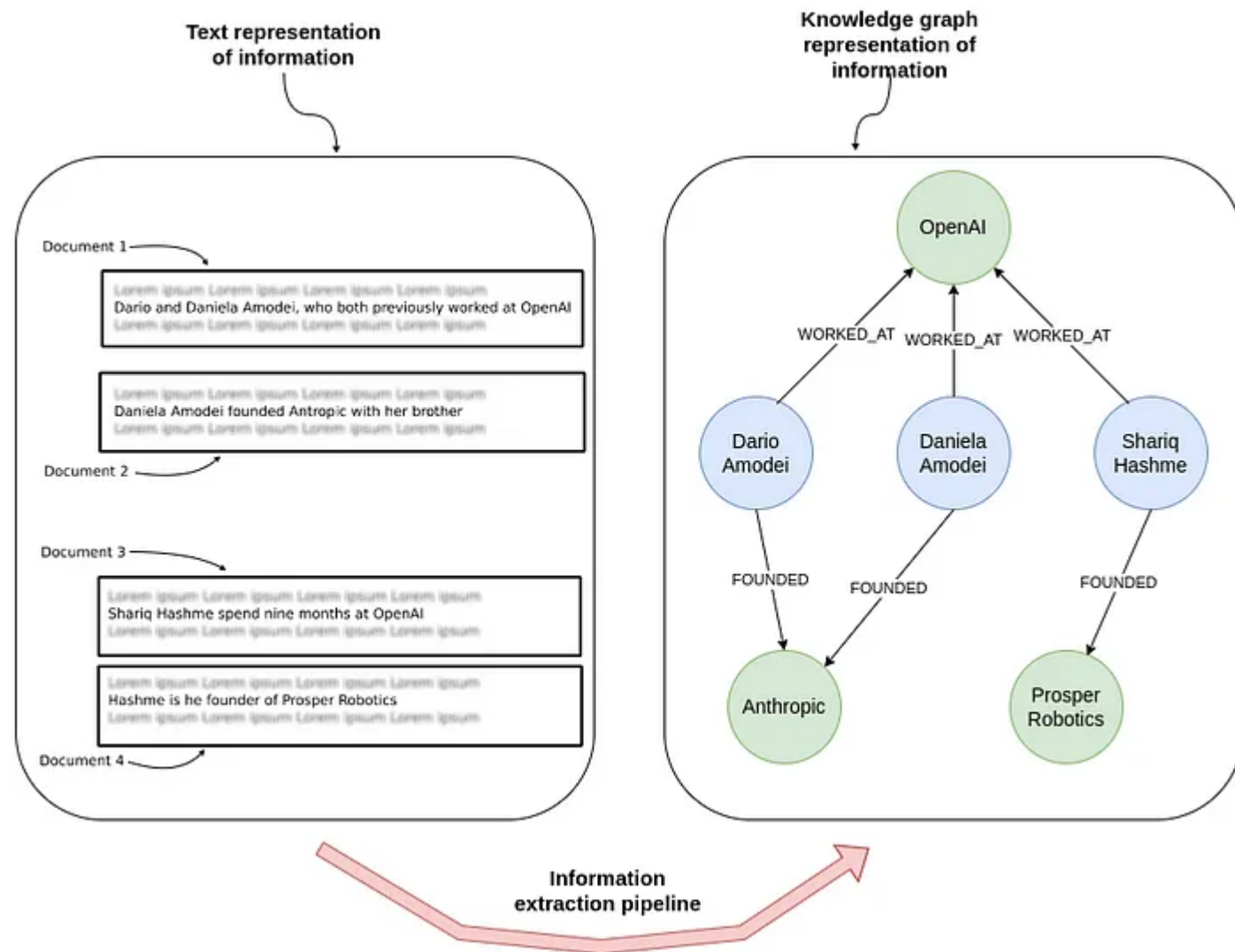
- **LLMs** têm uma forte capacidade de compreensão e geração de **linguagem natural**, mas às vezes podem **alucinar** fatos.
- **KGs** representam explicitamente o **conhecimento factual** em um formato estruturado, mas carecem de compreensão da linguagem.
- **Juntos**, os LLMs podem fornecer **contexto semântico**.



LLMs e Knowledge Graph

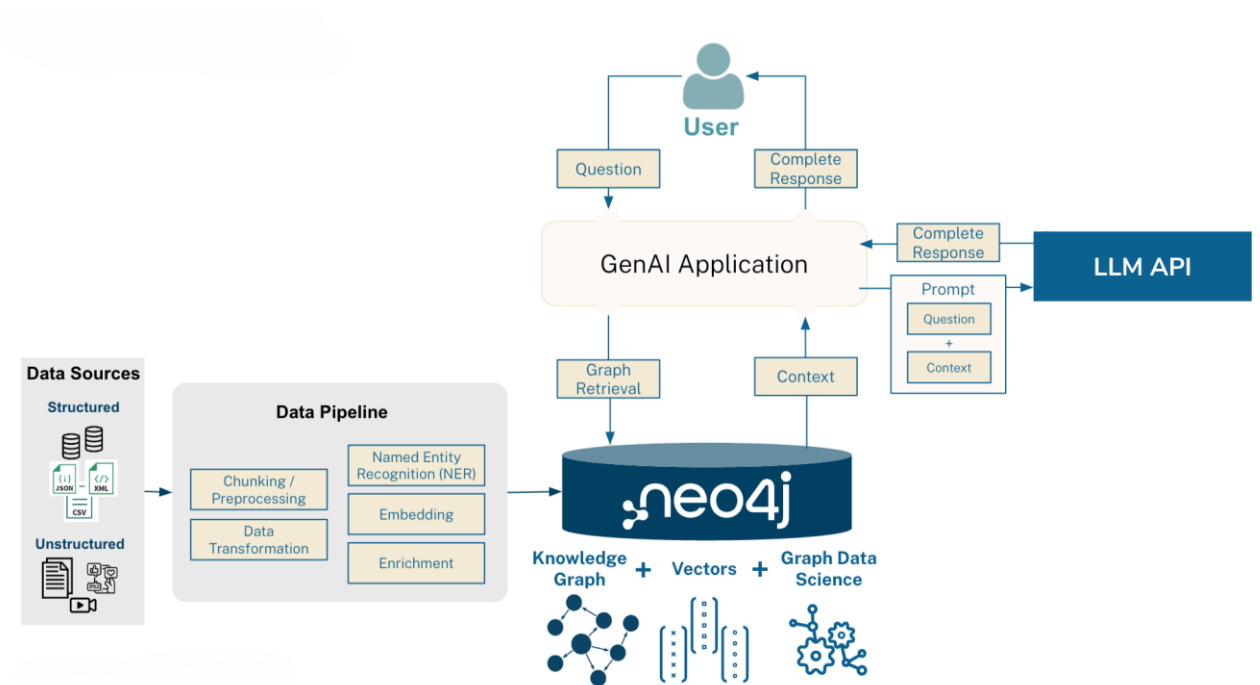
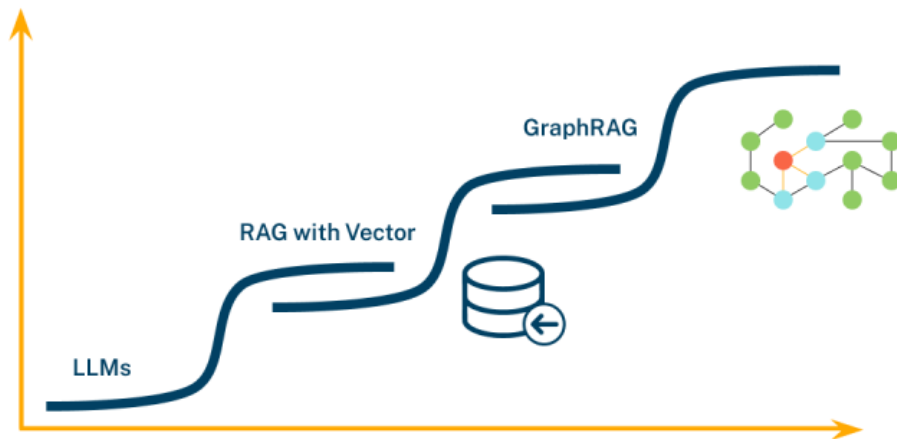
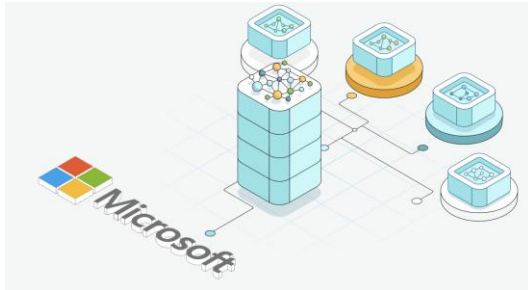


LLMs e Knowledge Graph



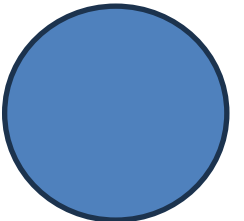
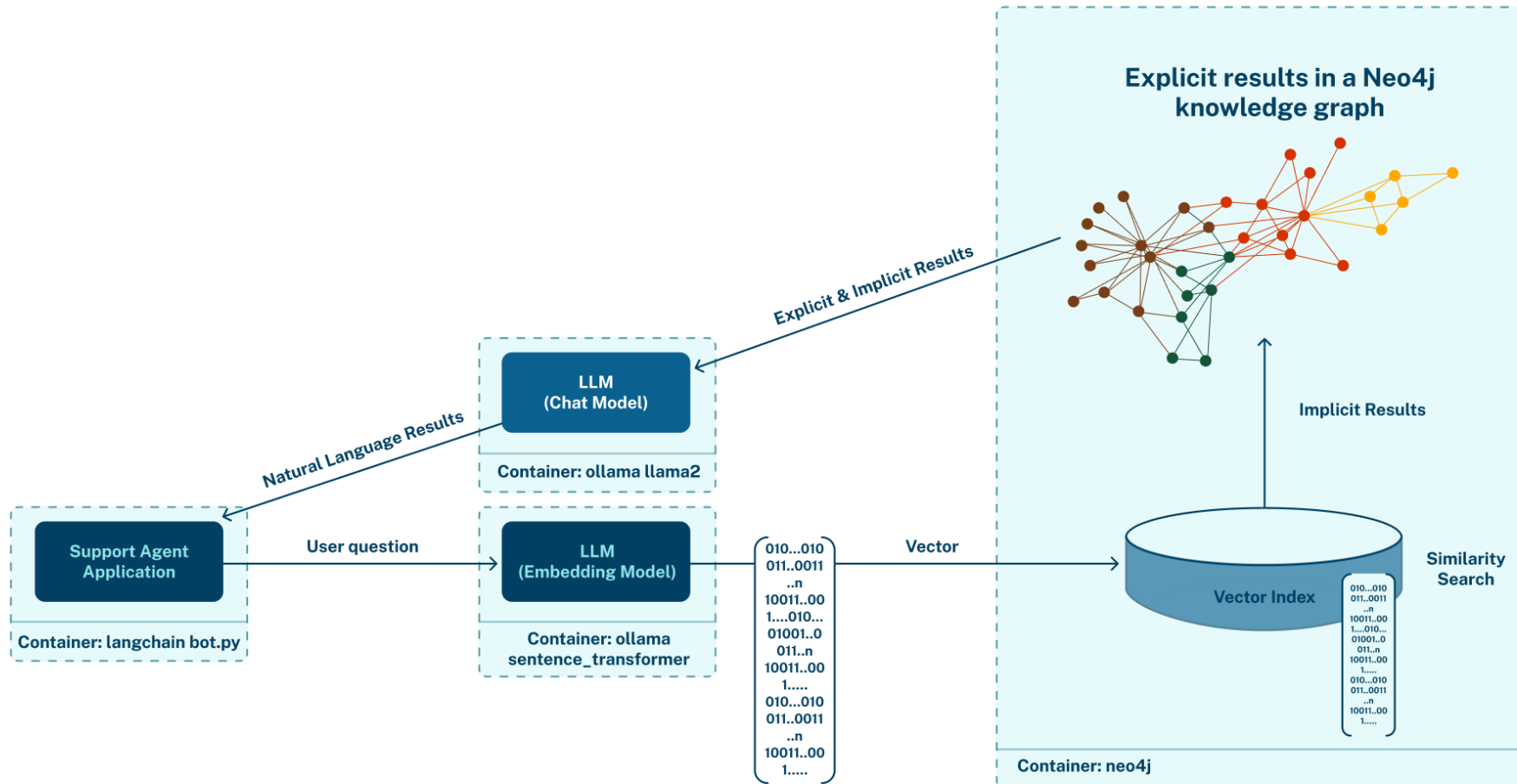
LLMs e Knowledge Graph

Microsoft Fabric + Neo4j



LLMs e Knowledge Graph

Neo4j



LLMs e Knowledge Graph

1. Criar uma instance no Neo4J AuraDB

neo4j aura / New Organization / New tenant

Get started
Developer hub

Data services

Instances

Graph Analytics
Data APIs

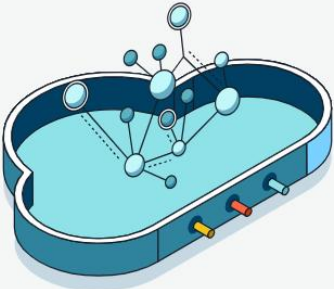
Tools

Import
Query
Explore
Dashboards

Operations
Project
Learning

Instances

Aura (0) Self-managed (0)



Create your first instance

Configure your instance and start realizing the endless possibilities with graphs.

Create instance

Have data? Import existing data into Neo4j.

Free

Learn and explore graphs for free

- ✓ Up to 200k nodes and 400k relationships
- ✗ Limited memory and CPU
- ✗ Limited backups
- ✗ Auto-deleted after 30 days of inactivity

\$0 GB/hour

Aguardar a criação.

LLMs e Knowledge Graph

1. Criar uma instance no **Neo4J AuraDB**

Instance01
● **RUNNING**
ID: 2a6100b4

Type: AuraDB Free Nodes: 0 (0%) Relationships: 0 (0%)

> Metrics

- Query
- Explore
- Dashboards
- Developer hub

Connect ^ ...

Upgrade

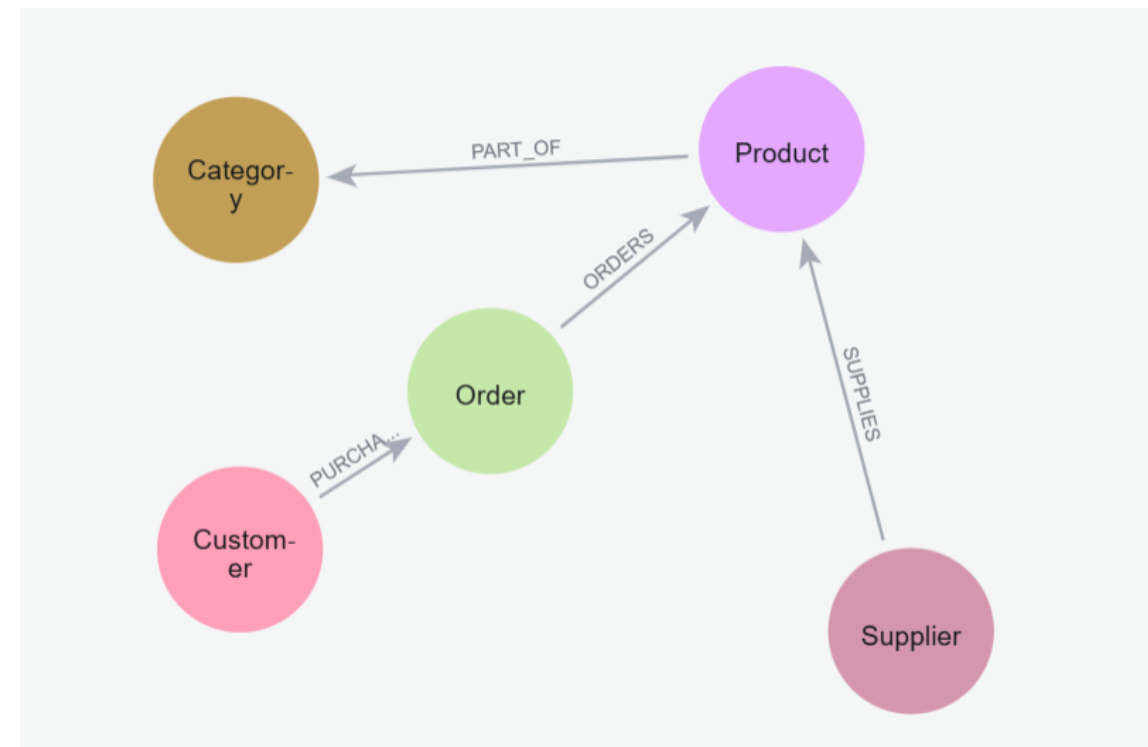
LLMs e Knowledge Graph

2. Popular a base de dados (script vendas.txt)

Vendas.txt

The screenshot displays the Neo4j Aura web interface. On the left, a sidebar contains navigation links for 'Get started', 'Developer hub', 'Data services', 'Instances', 'Graph Analytics', 'Data APIs', 'Tools', 'Import', 'Query', 'Explore', 'Dashboards', 'Operations', 'Project', and 'Learning'. The main area is titled 'Database information' and shows 'Nodes (1,035)' and 'Relationships (3,139)'. Below this, there are tabs for 'ORDERS', 'PART_OF', 'PURCHASED', and 'SUPPLIES'. A 'Property keys' section lists various attributes like 'address', 'categoryID', 'categoryName', 'city', 'companyName', 'contactName', 'contactTitle', 'country', 'customerID', 'description', 'discontinued', 'discount', 'employeeID', 'fax', 'freight', 'homePage', 'orderDate', 'orderID', 'phone', and 'picture'. On the right, a query editor shows a series of Cypher commands: creating constraints for Product, Category, Supplier, Customer, and Order, followed by two 'LOAD CSV' statements for product and category data.

CALL db.schema.visualization()



3a. Explorar a base de dados

```
MATCH (p:Product)-[rel:PART_OF]
      ->(c:Category {categoryName: 'Beverages'})
RETURN p, rel, c;
```

```
MATCH (c:Customer)-[r1:PURCHASED]->
      (o:Order)-[r2:ORDERS] ->
      (p:Product{productName: 'Ipoh Coffee'})
RETURN c.companyName, COUNT(o) AS orders, collect(o.orderID)
ORDER BY orders DESC
```



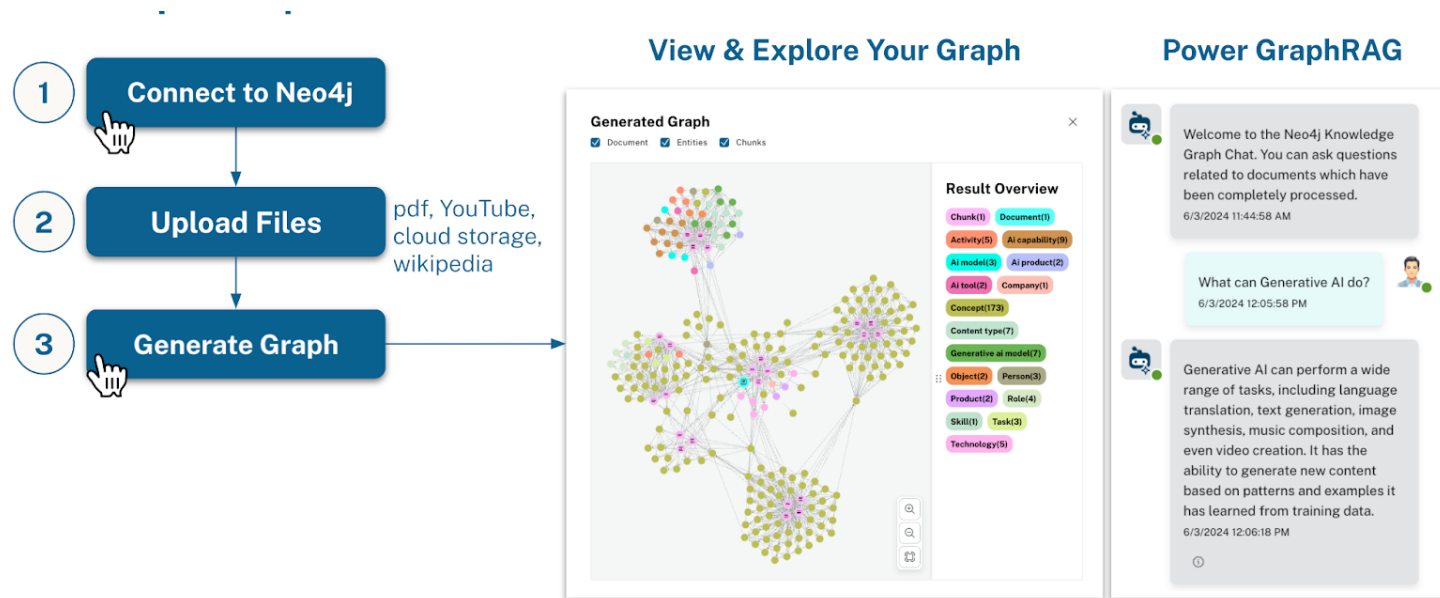
LLMs e Knowledge Graph

3b. Explorar a base de dados

```
MATCH (cust:Customer)-[r1:PURCHASED] ->
  (o:Order)-[r2:ORDERS]->
  (p:Product)-[r3:PART_OF]->
  (c:Category{categoryName:'Produce'}), (p)<-
  [r4:SUPPLIES]-(s:Supplier)
RETURN *;
```

LLMs e Knowledge Graph

4) Neo4J GenAI – Integrar LLM



- Inserir dados não estruturados na base existente a partir de documentos pdf
- Acessar Neo4j LLM Knowledge Graph Builder - Extract Nodes and Relationships from Unstructured Text

LLMs e Knowledge Graph

FIAP

3) Neo4J GenAI – Integrar LLM

llm-graph-builder.neo4j.com

neo4j

Connect to Neo4j to upload documents

Neo4j connection

Not Connected

Graph Schema configured(5 Labels + 4 Rel Types)

Graph Settings Connect to Neo4j

☐	Name	Status	Upload Status	Size (KB)	Source
--------------------------	------	--------	---------------	-----------	--------

Showing 0 of 0 results

LLM Model for Processing & Chat

Gemini 2.5 flash

Generate Graph Delete Files Preview Graph

Welcome to the Neo4j Knowledge Graph Chat. You can ask questions related to documents which have been completely processed.
18/02/2026 16:38:32

Connect to Neo4j

Don't have a Neo4j instance? Start for free today

Drop your neo4j credentials file here
or [browse](#)

Protocol bolt URI 34.238.28.20:7687

Database neo4j

Username neo4j Password

Connect

1

2

3

3) Neo4J GenAI – Integrar LLM

Neo4j connection

neo4j+s://2a6100b4.databases.neo4j.io:7687 / neo4j

No Graph Schema configured

	Name	Status	Upload Status	Size (KB)	Source	Type	Model	Nodes	Relations
--	------	--------	---------------	-----------	--------	------	-------	-------	-----------

Enhance graph quality

Graph Settings

Disconnect

Graph Settings

This set of tools will help you enhance the quality of your Knowledge Graph by removing possible duplicated entities, disconnected nodes and set a Graph Schema for improving the quality of the entity extraction process

Entity Extraction Settings

Processing Configuration

Disconnected Nodes

De-Duplication Of Nodes

Post Processing Jobs

1.Predefine the structure of your knowledge graph by selecting specific node and relationship labels.

2.Focus your analysis by extracting only the relationships and entities that matter most to your use case. Achieve a cleaner and more insightful graph representation tailored to your domain.

Graph Pattern

Select/Create Source

Select...

Select/Create Type

Select...

Select/Create Target

Select...

+ Add

Selected Patterns

Predefined Schema

Load Existing Schema

Get Schema From Text

Data Importer JSON

Add Schema from ...

Clear Schema

Apply

Gemini 2.5 flash

Generate Graph

Delete Files

Preview Graph

3) Neo4J GenAI – Integrar LLM

Schema From Database



Selected Patterns

5 Nodes & 4 Relationships

Order -[:ORDERS]-> Product X

Customer -[:PURCHASED]-> Order X

Product -[:PART_OF]-> Category X

Supplier -[:SUPPLIES]-> Product X

Cancel

Apply



LLMs e Knowledge Graph

3) Neo4J GenAI – Integrar LLM

The screenshot displays the Neo4j GenAI interface. On the left, a 'Drag & Drop' area allows users to upload documents, images, or unstructured text. Below this are options for 'Web Sources' and 'Amazon S3'. The main panel shows a 'Neo4j connection' status with a table header for 'Name', 'Status', 'Upload Status', 'Size (KB)', and 'Source'. A 'Web Sources' modal dialog is open, featuring a play button, a 'W' logo, and a globe icon. It includes a 'Website Link' input field with the text 'https://neo4j.com/' and 'Reset' and 'Submit' buttons. A right sidebar contains a welcome message and a date/time stamp. The bottom of the interface includes an 'LLM Model for Processing & Chat' dropdown set to 'Gemini 2.5 flash', and buttons for 'Generate Graph', 'Delete Files', and 'Preview Graph'. An 'Ask' button is located in the bottom right corner.

Neo4j connection
neo4j+s://2a6100b4.databases.neo4j.io:7687 / neo4j
● Graph Schema configured(5 Labels + 4 Rel Types)

Web Sources
Convert Any Web Source to Knowledge graph

Website Link
https://neo4j.com/

Reset Submit

LLM Model for Processing & Chat
Gemini 2.5 flash

Generate Graph Delete Files Preview Graph

Welcome to the Neo4j Knowledge Graph Chat. You can ask questions related to documents which have been completely processed.
18/02/2026 16:56:30

Ask

LLMs e Knowledge Graph



Neo4j connection

neo4j+s://2a6100b4.databases.neo4j.io:7687 / neo4j

Graph Settings

Disconnect

● Graph Schema configured(5 Labels + 4 Rel Types)

☐	Name	Status ▾	Upload Status	Size (KB)	Source ▾	Type ▾	Model ▾
☐	developers-gui...	▲ Processing ✕	● Uploaded	7.81	web-url ↗	TEXT	Gemini 2.5 flash
☐	developers-gui...	▲ Processing ✕	● Uploaded	7.81	web-url ↗	TEXT	Gemini 2.5 flash



FIAP

Ipoh Coffee — Qtde: 6 — Preço unitário: \$46.00 — Total: \$276.00
Pavlova — Qtde: 3 — Preço unitário: \$17.45 — Total: \$52.35
Singaporean Hokkien Fried Mee — Qtde: 6 — Preço unitário: \$14.00 — Total: \$84.00
Totais: quantidade 15; soma dos preços unitários \$77.45; total geral \$412.35.

Details



To generate this response, the process took **26.17 seconds**,utilizing **3232** tokens with the model **gpt-5-mini** in **graph & vector & fulltext** mode.

Evaluation Metrics

Customer (1) Wilman Kala

Graph Entities used for Answer Generation



LLMs e Knowledge Graph

3b. Associar Produto a Entity na base de dados

```
MATCH (p:Product) SET p.id=p.productName, p:__Entity__
```

3a. Explorar a base de dados

```
MATCH (c:Customer)-[r1:PURCHASED]->
  (o:Order)-[r2:ORDERS] ->
  (p:Product{productName: 'Filo Mix'})
RETURN c.companyName, COUNT(o) AS orders, collect(o.orderID)
ORDER BY orders DESC
```

LLMs e Knowledge Graph



Exercício

- Modelagem e Ingestão de Dados
- Análise

Airbnb – Análise de uma cidade



LLMs e Knowledge Graph



Exercício

Atividades

1. Modelagem e Ingestão de dados – base de dados grafos
2. Explorar algoritmos ML com grafos

· LLMs e Knowledge Graph



· Exercício

Ferramentas

1. **Neo4J Sandbox** – banco de dados - <https://sandbox.neo4j.com/>
2. Modelar e importar dados – <https://data-importer.neo4j.io/>
3. Neo4J Bloom – realizar analyses GDS

LLMs e Knowledge Graph



Exercício

1. Dados Airbnb

- a. Arquivos disponíveis em <https://insideairbnb.com/get-the-data/>
- b. Escolher uma cidade
(regra: dois ou mais grupos não podem explorar a mesma cidade)

LLMs e Knowledge Graph



Exercício

2. Modelagem e Ingestão de dados

Airbnb – criar base de dados

- Baixar arquivos listing.csv (detalhado) e reviews.csv

<https://insideairbnb.com/get-the-data/>

Rio de Janeiro, Rio de Janeiro, Brazil

27 June, 2024 (Explore)

Country/City	File Name	Description
Rio de Janeiro	listings.csv.gz	Detailed Listings data
Rio de Janeiro	calendar.csv.gz	Detailed Calendar Data
Rio de Janeiro	reviews.csv.gz	Detailed Review Data
Rio de Janeiro	listings.csv	Summary information and metrics for listings in Rio de Janeiro (good for visualisations).
Rio de Janeiro	reviews.csv	Summary Review data and Listing ID (to facilitate time based analytics and visualisations linked to a listing).
Rio de Janeiro	neighbourhoods.csv	Neighbourhood list for geo filter. Sourced from city or open source GIS files.
Rio de Janeiro	neighbourhoods.geojson	GeoJSON file of neighbourhoods of the city.

[show](#) archived data

(generally quarterly data for the last 12 months. For additional data, make an archived [data request](#).)

LLMs e Knowledge Graph



Exercício

2. Modelagem e Ingestão de dados

Airbnb – criar base de dados

- Acessar Neo4J Sandbox
- Criar um projeto Blank Sandbox - Graph Data Science

Exercício

2. Modelagem e Ingestão de dados

New Organization

New tenant

Feedback

Graph models / listings model 1

Generate model

Run import

Data source

Local

Browse

listings.csv

- id 8211980...
- name Bright a...
- host_id 84350716
- host_na... Katrina
- neighbo...
- neighbo... Leblon
- latitude -22.982...
- longitude -43.222...
- room_ty... Entire ho...
- price 580
- minimu... 2
- number_... 86
- last_revi... 2025-09...
- reviews_... 2.72
- calculat... 1

```

graph TD
    Listing((Listing)) -- possible --> Room_type((Room_type))
    Listing -- file --> Neighbourhood((Neighbourhood))
    Listing -- has --> Host((Host))
  
```

Definition

Constraints & Indexes (2)

Label

Listing

Table

listings.csv

Filter table

Properties

Map from table

☐	Name	Type	Column
☐	id	integer	id

ID

A property that uniquely identifies the node.

id

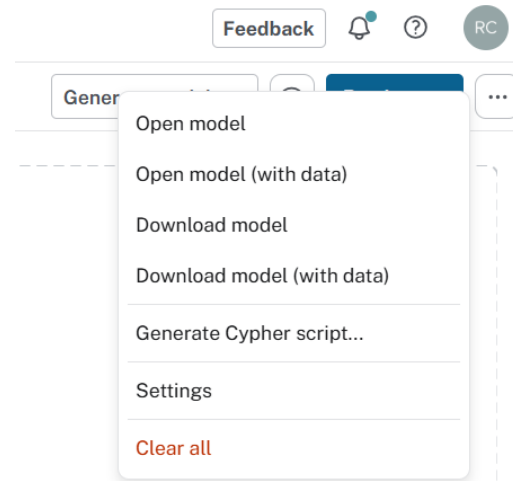
LLMs e Knowledge Graph

Exercício

2. Modelagem e Ingestão de dados

Airbnb – modelo grafos

- Salvar modelo – três pontinhos – Download Model
- Run Import



Import results

Total time: 00:13:54

✓ **Run import completed**
Import completed successfully.

✓ Listing	listings.csv						Show Cypher
Time taken	File size	File rows	Nodes created	Properties set	Labels added	Query time	
00:00:46	66.8 MiB	34,664	34,664	344,690	34,664	00:00:41	

✓ Host	listings.csv						Show Cypher
Time taken	File size	File rows	Nodes created	Properties set	Labels added	Query time	
00:00:39	66.8 MiB	34,664	20,772	167,350	20,772	00:00:35	

✓ Review	reviews.csv						Show Cypher
----------	-------------	--	--	--	--	--	-------------

Close

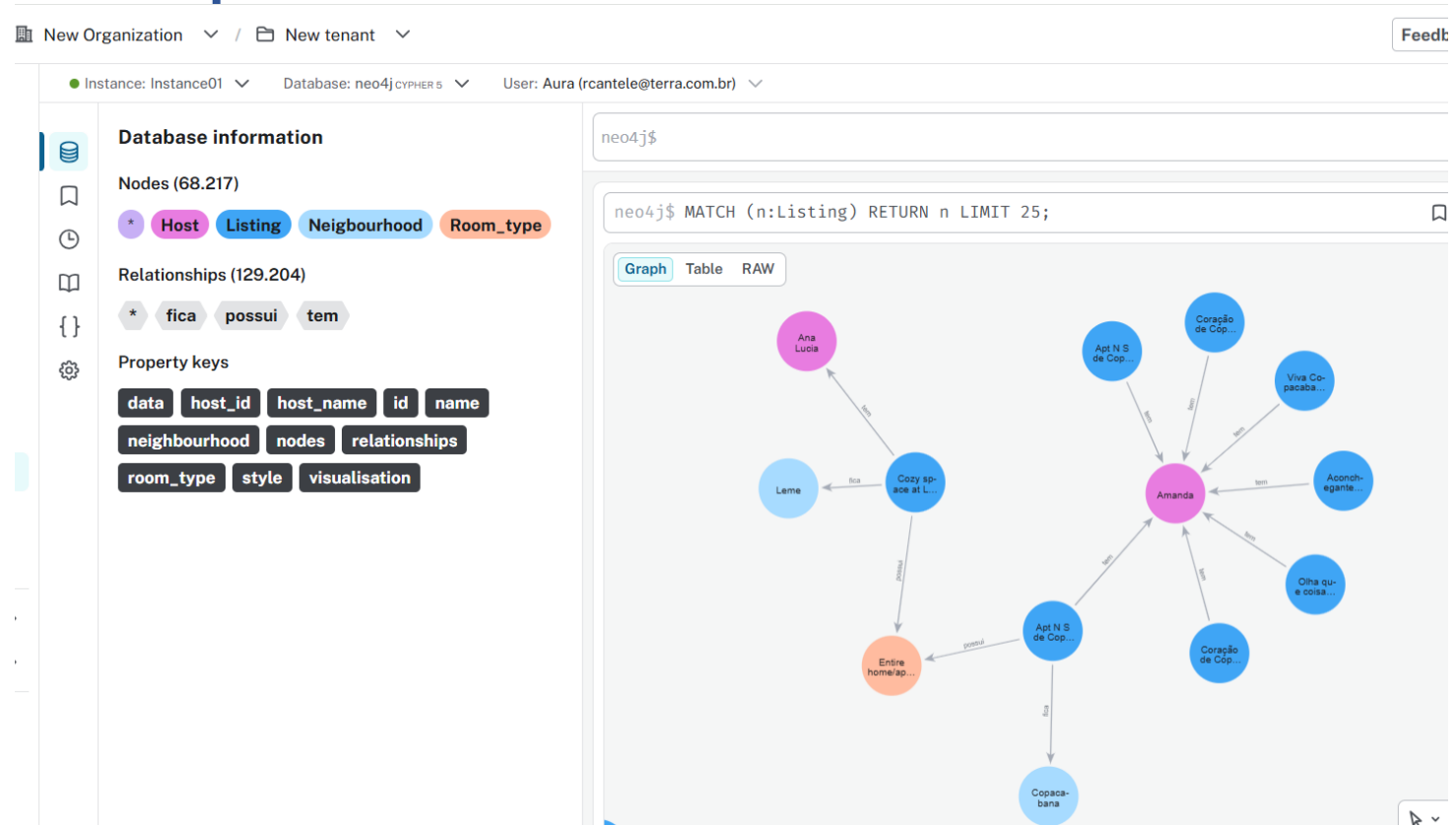
Query results

LLMs e Knowledge Graph

Exercício

3. Explorar algoritmos ML com grafos

Browser / Explorar Host com vários Listing

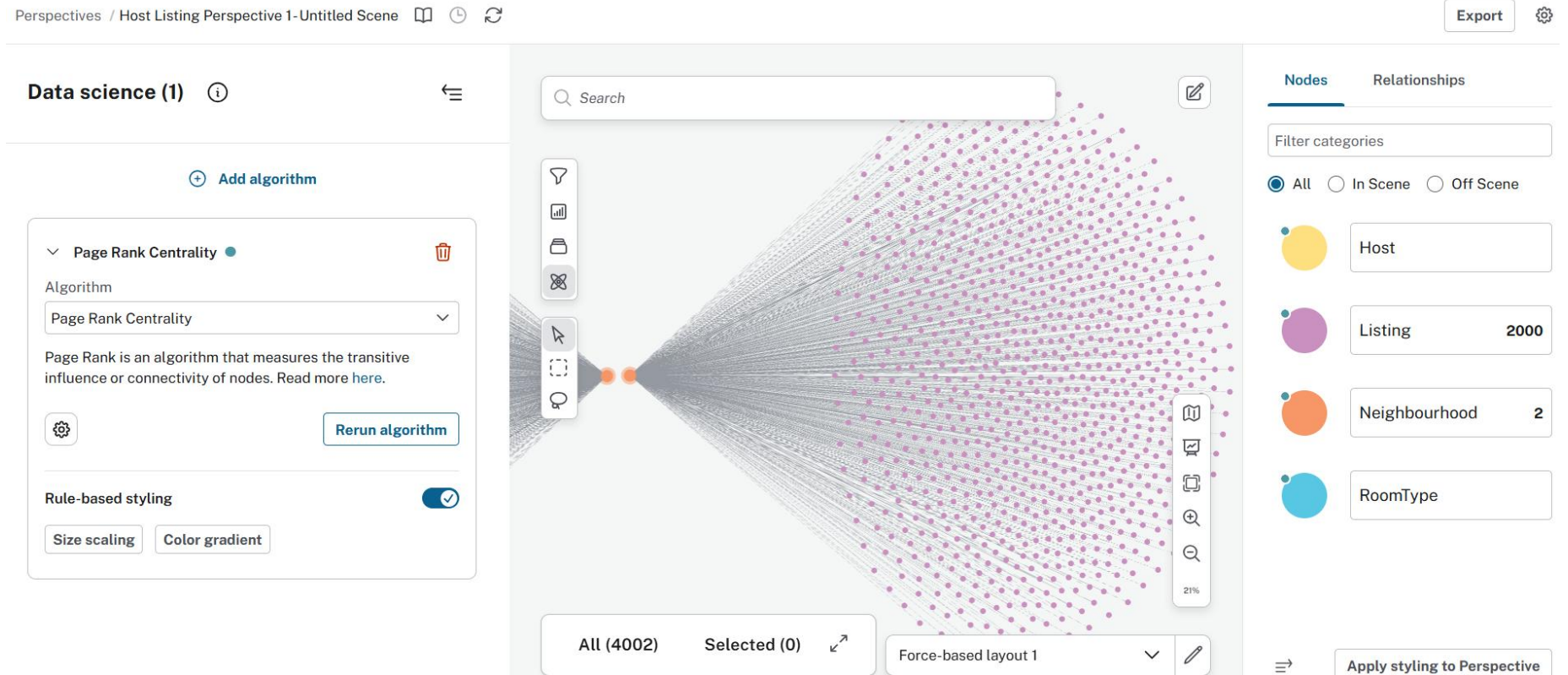


LLMs e Knowledge Graph

Exercício

3. Explorar algoritmos ML com grafos

Neo4J Bloom / Pesquisar e realizar GDS



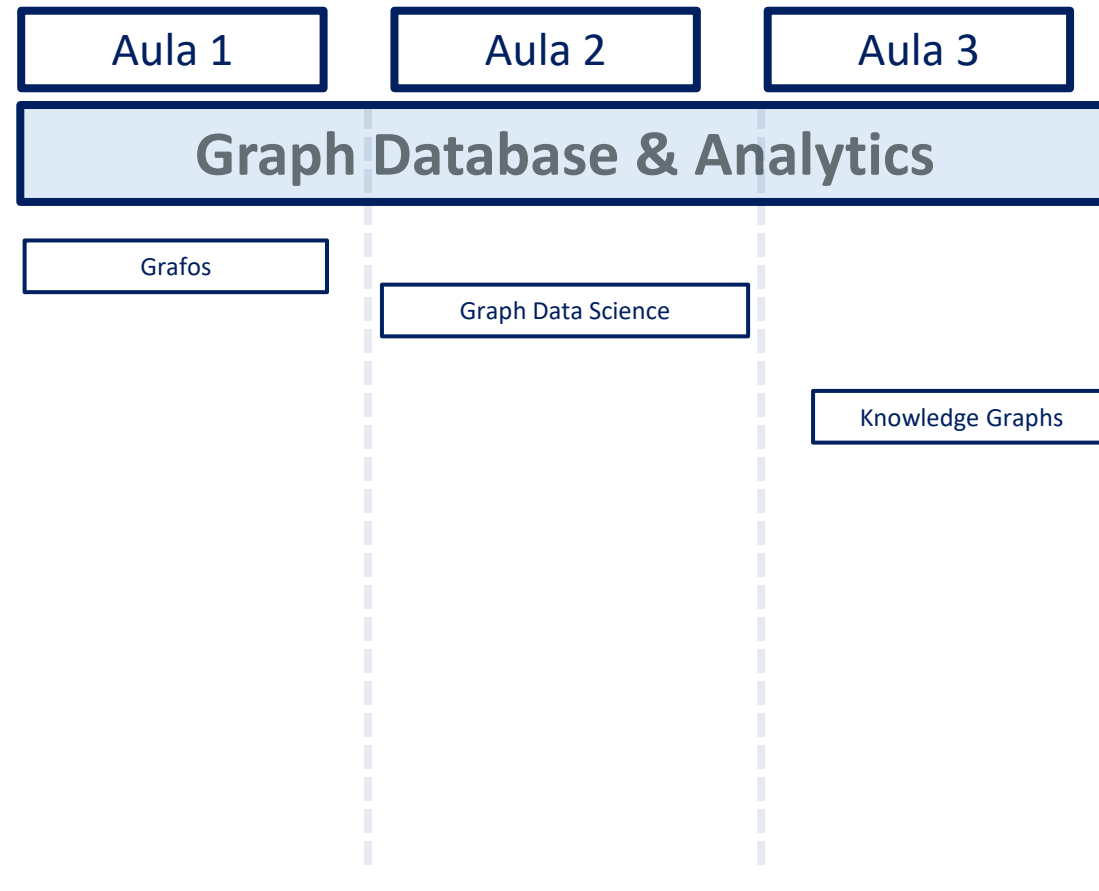
LLMs e Knowledge Graph



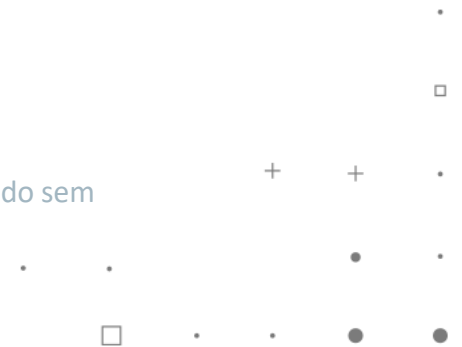
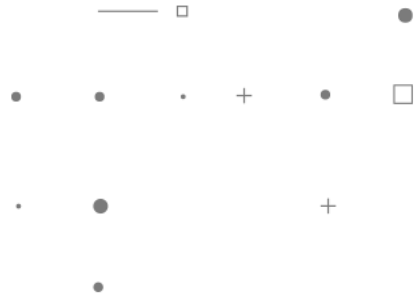
Exercício

- Entrega de evidências da realização das atividades.
- Apresentar em aula os resultados

Resumo







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